

BFS Traversal

Step 1: Start at E

Legend

Complete

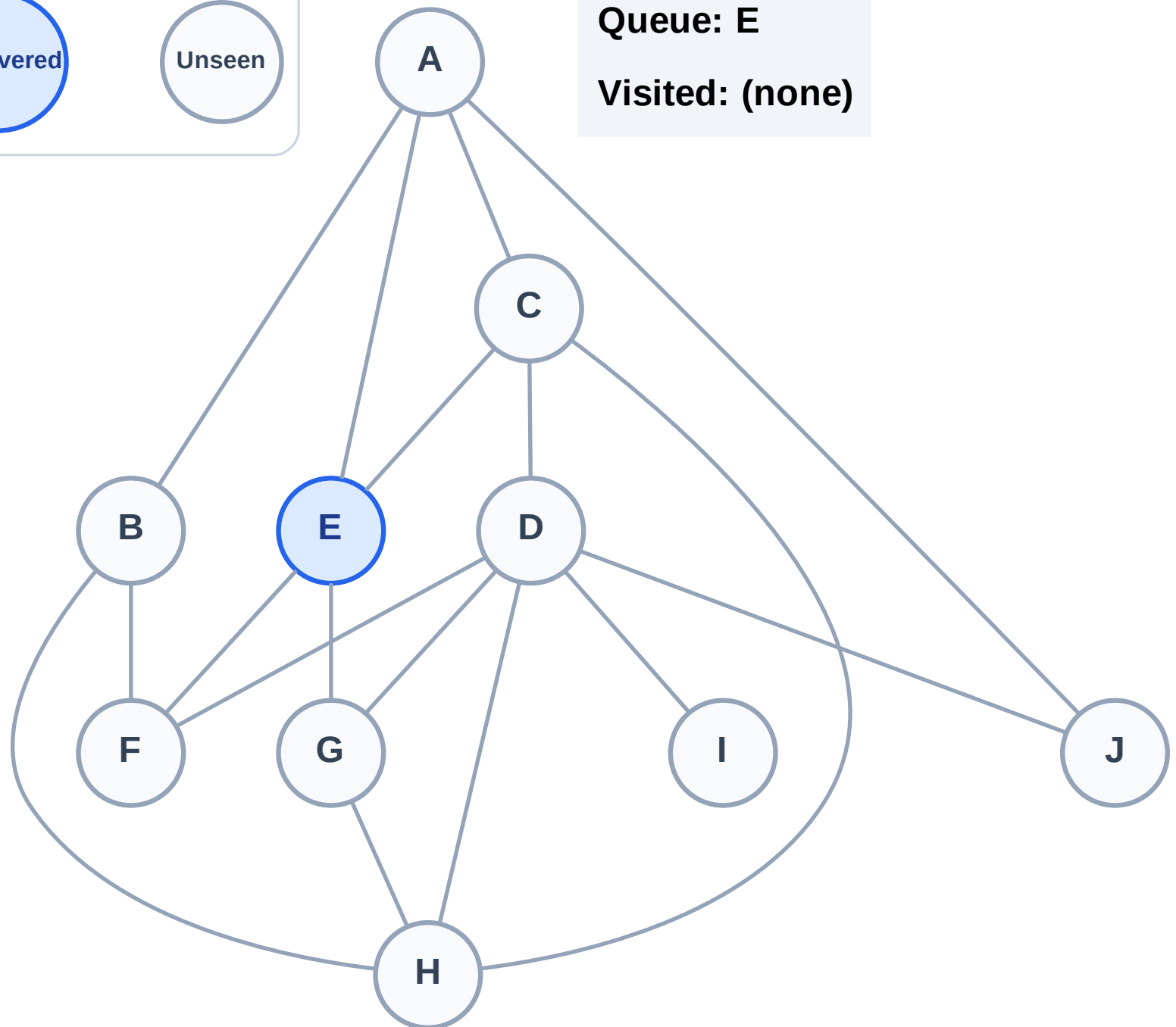
Processing

Discovered

Unseen

Queue: E

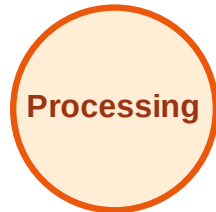
Visited: (none)



BFS Traversal

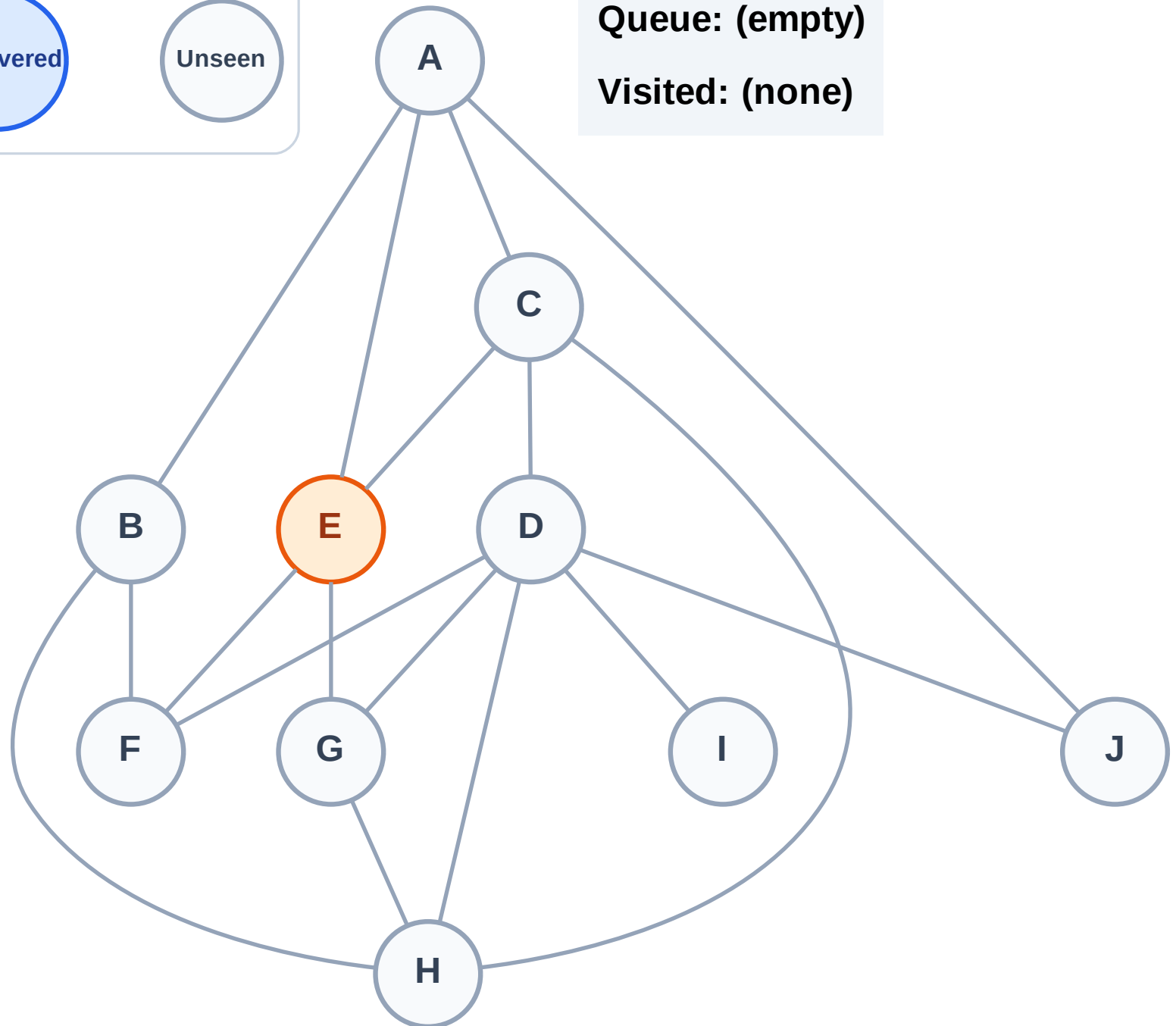
Step 2: Process node E

Legend



Queue: (empty)

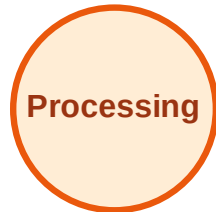
Visited: (none)



BFS Traversal

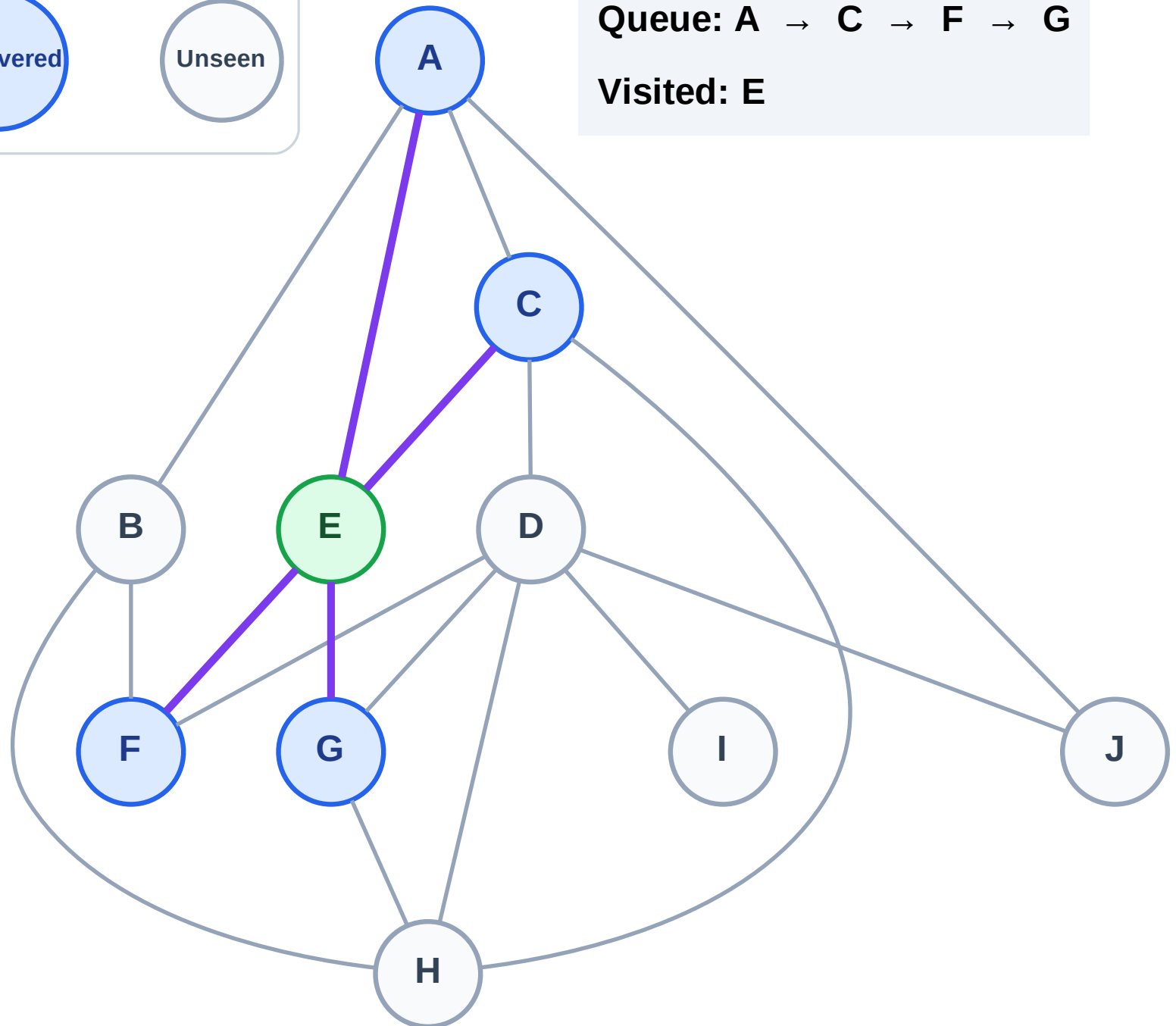
Step 3: Discover A, C, F, G from E

Legend



Queue: A → C → F → G

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

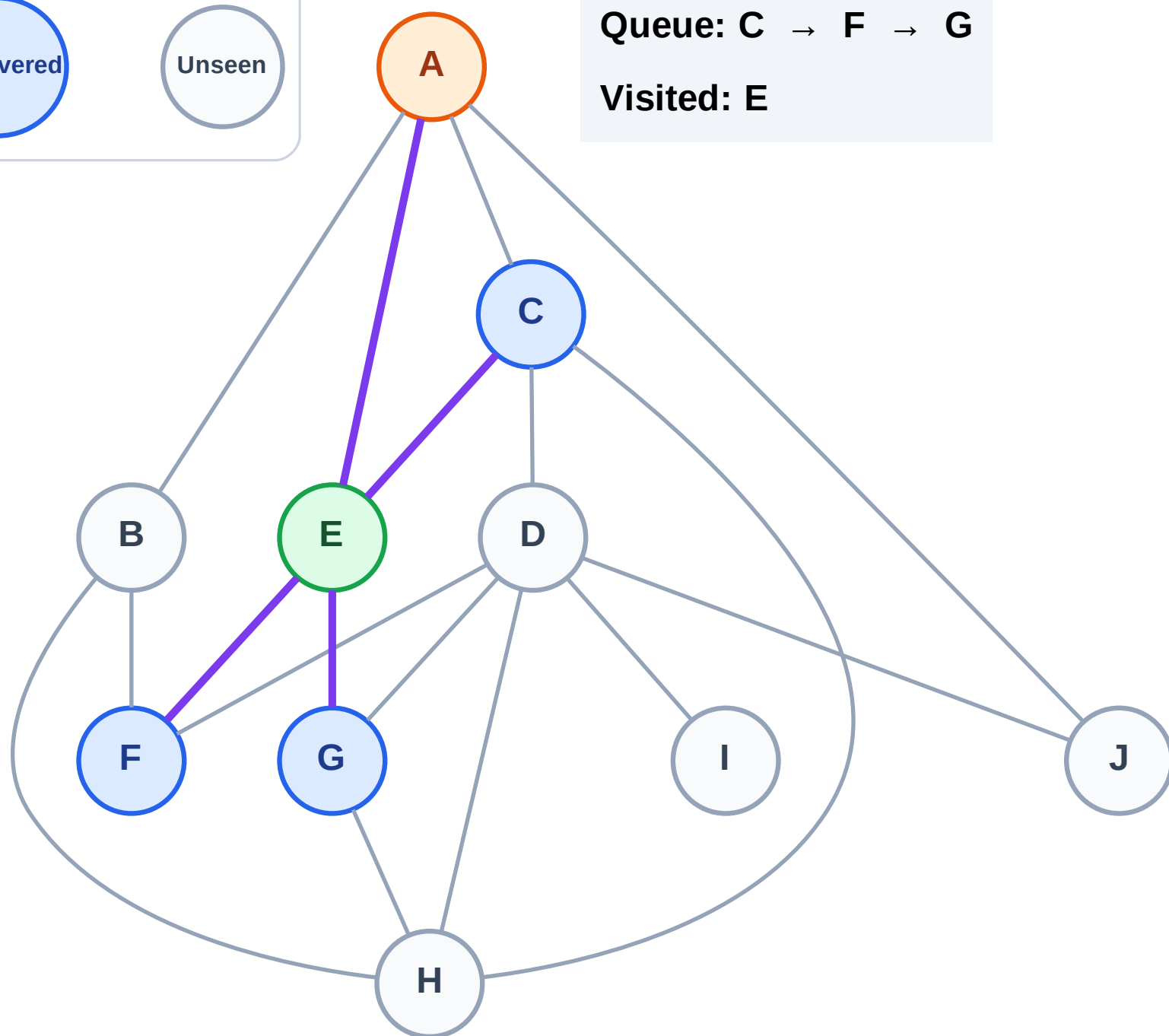
Processing

Discovered

Unseen

Queue: C → F → G

Visited: E



BFS Traversal

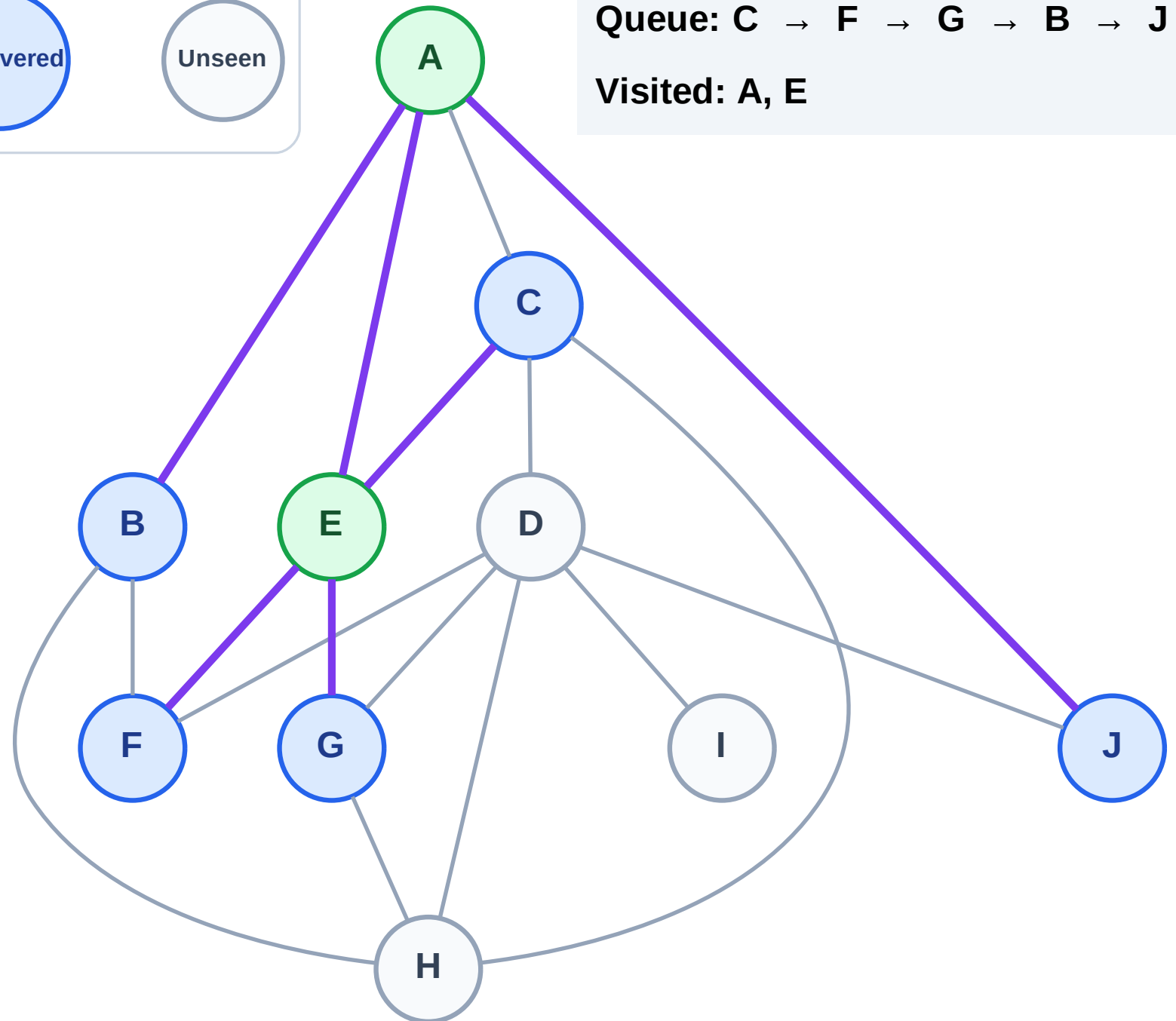
Step 5: Discover B, J from A

Legend



Queue: C → F → G → B → J

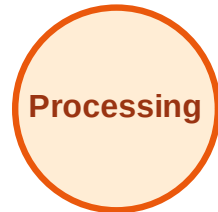
Visited: A, E



BFS Traversal

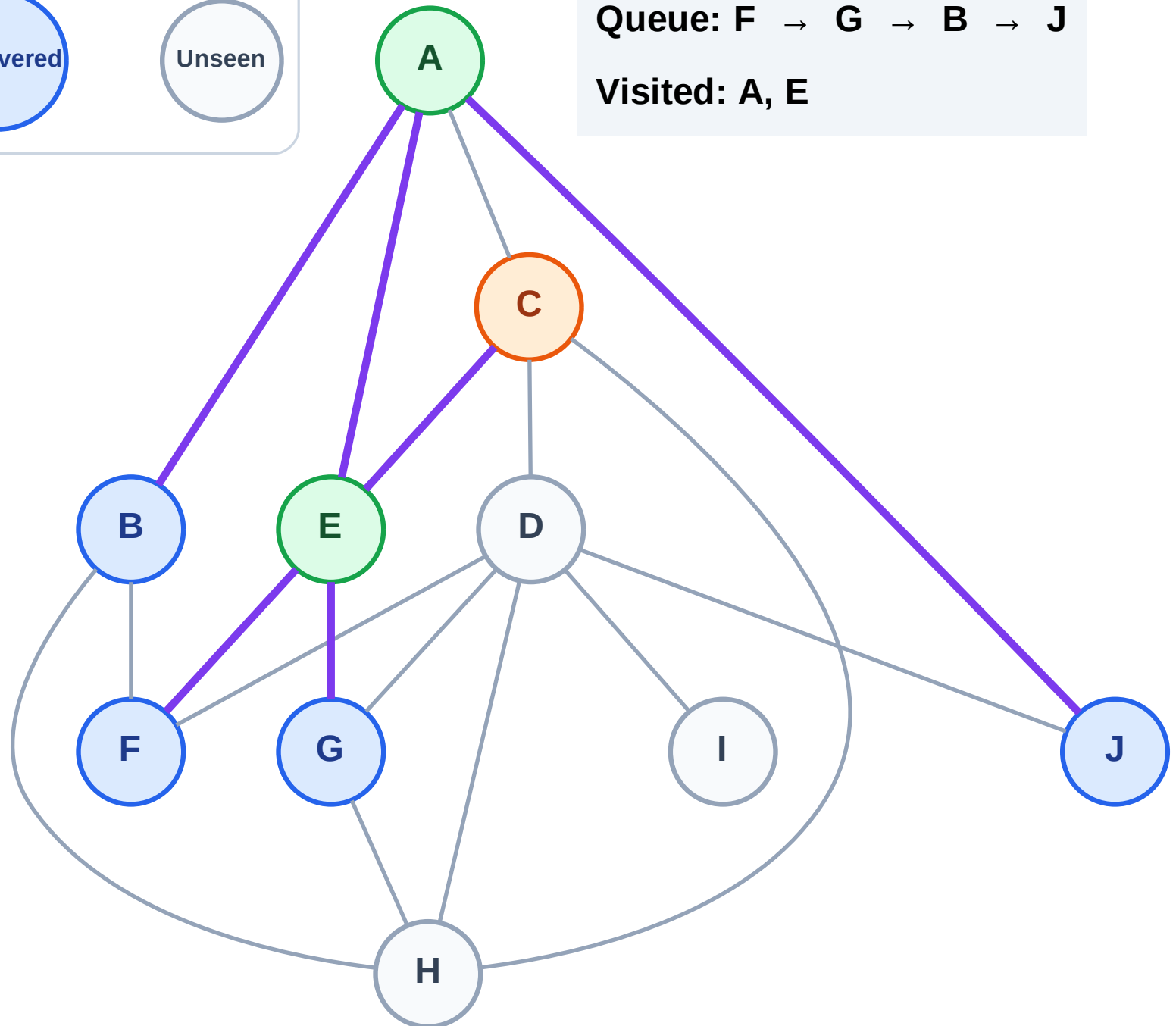
Step 6: Process node C

Legend



Queue: F → G → B → J

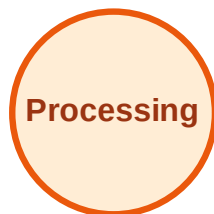
Visited: A, E



BFS Traversal

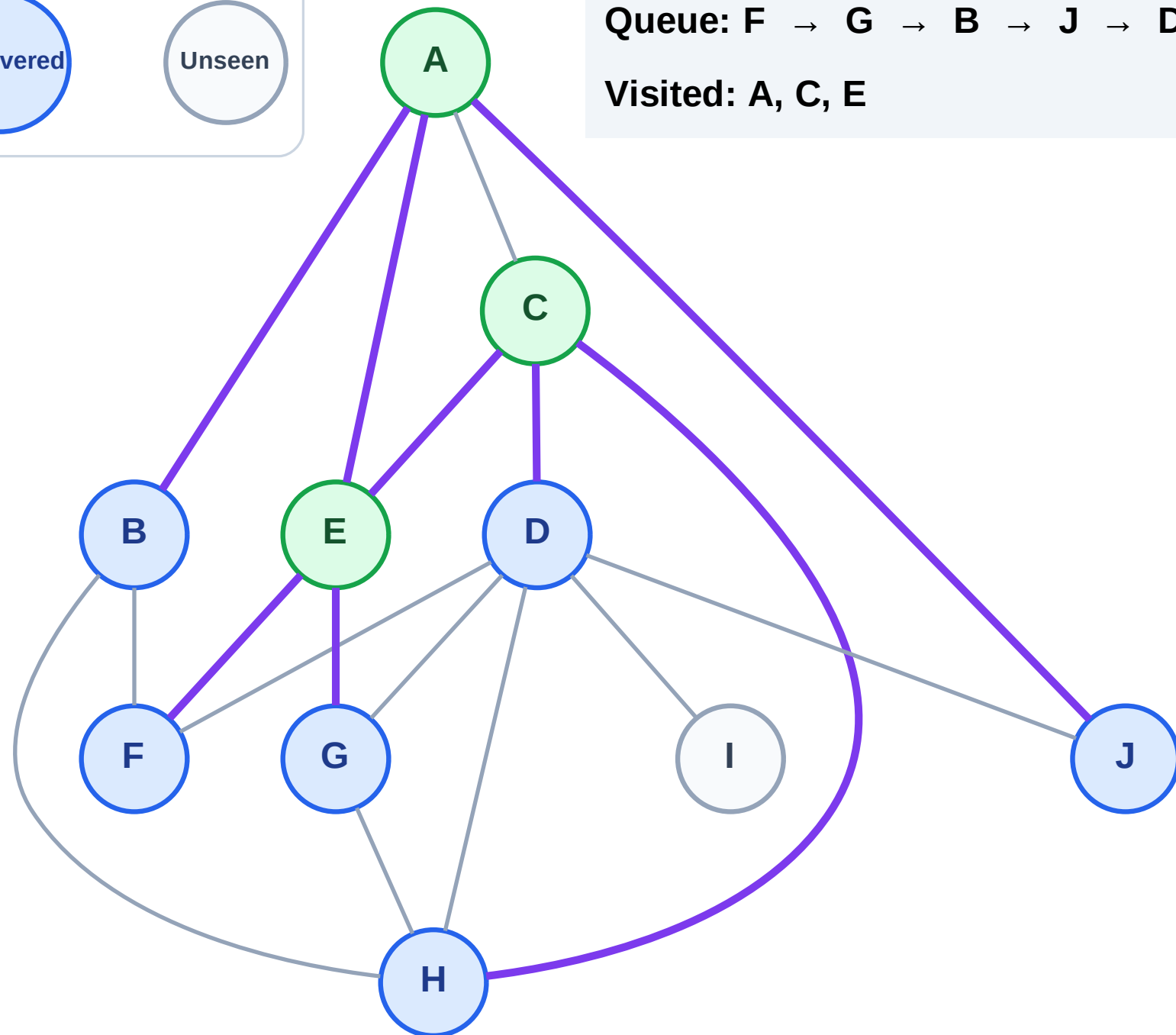
Step 7: Discover D, H from C

Legend



Queue: F → G → B → J → D → H

Visited: A, C, E



BFS Traversal

Step 8: Process node F

Legend

Complete

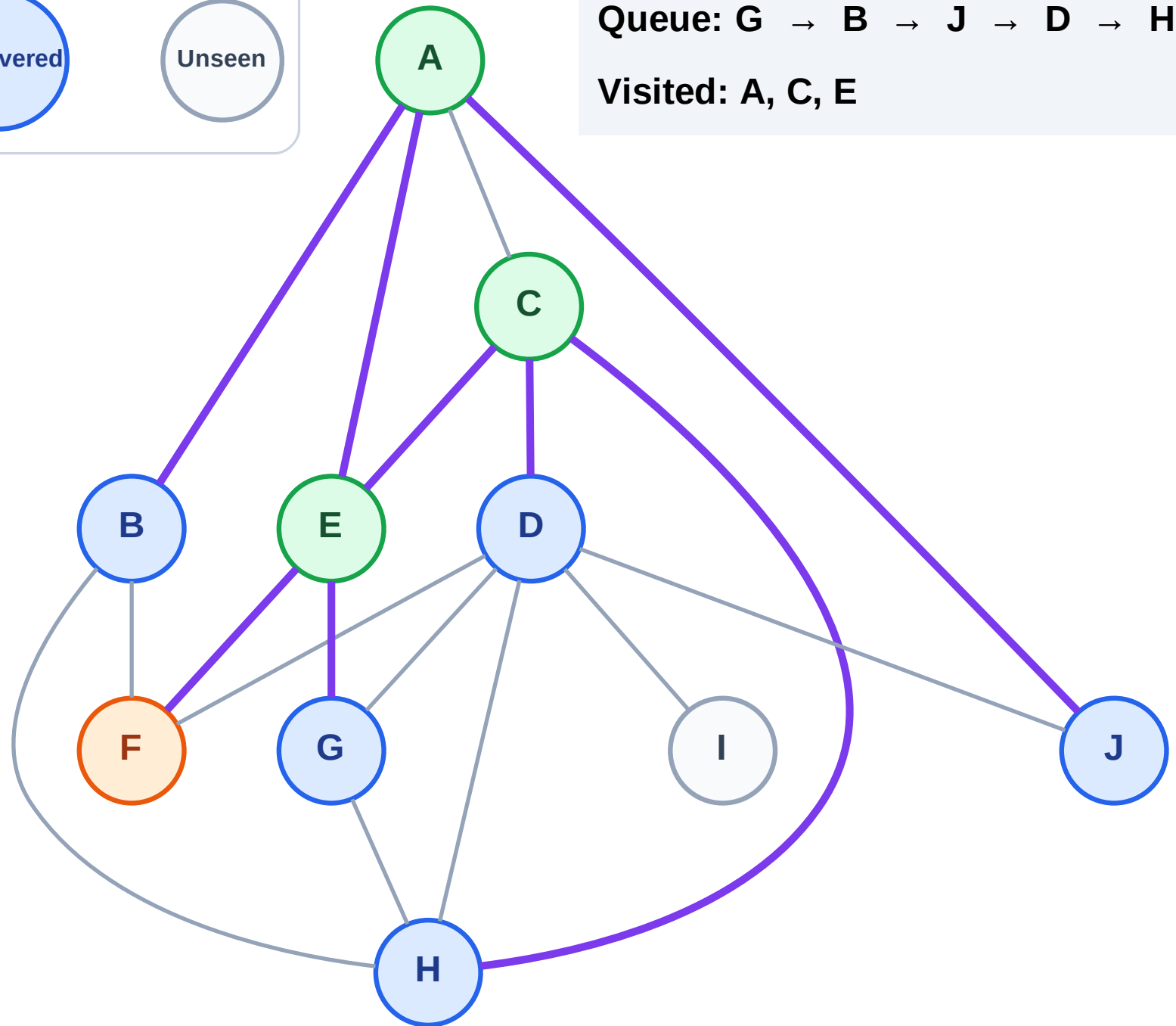
Processing

Discovered

Unseen

Queue: G → B → J → D → H

Visited: A, C, E



BFS Traversal

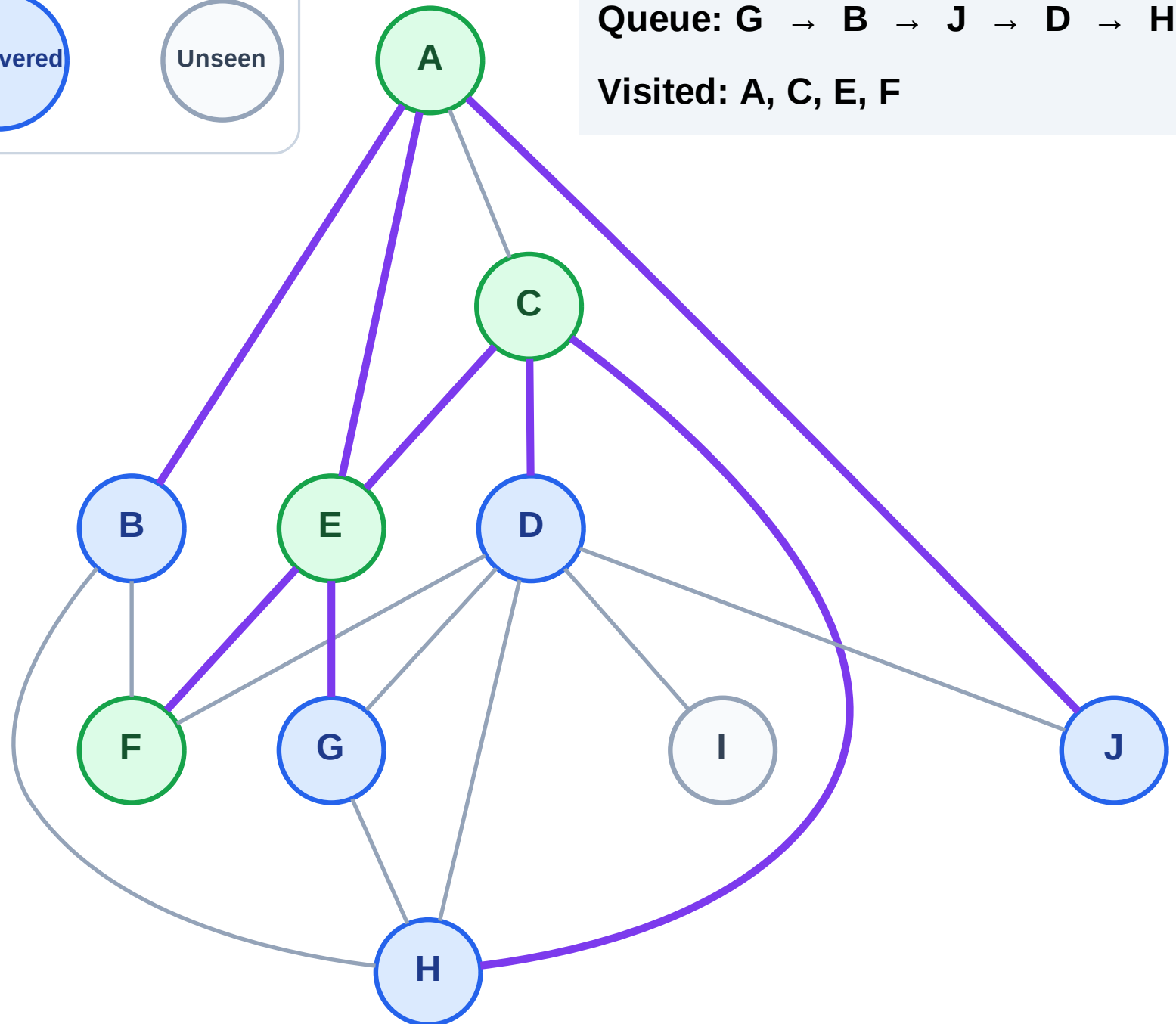
Step 9: No new neighbors from F

Legend



Queue: G → B → J → D → H

Visited: A, C, E, F



BFS Traversal

Step 10: Process node G

Legend

Complete

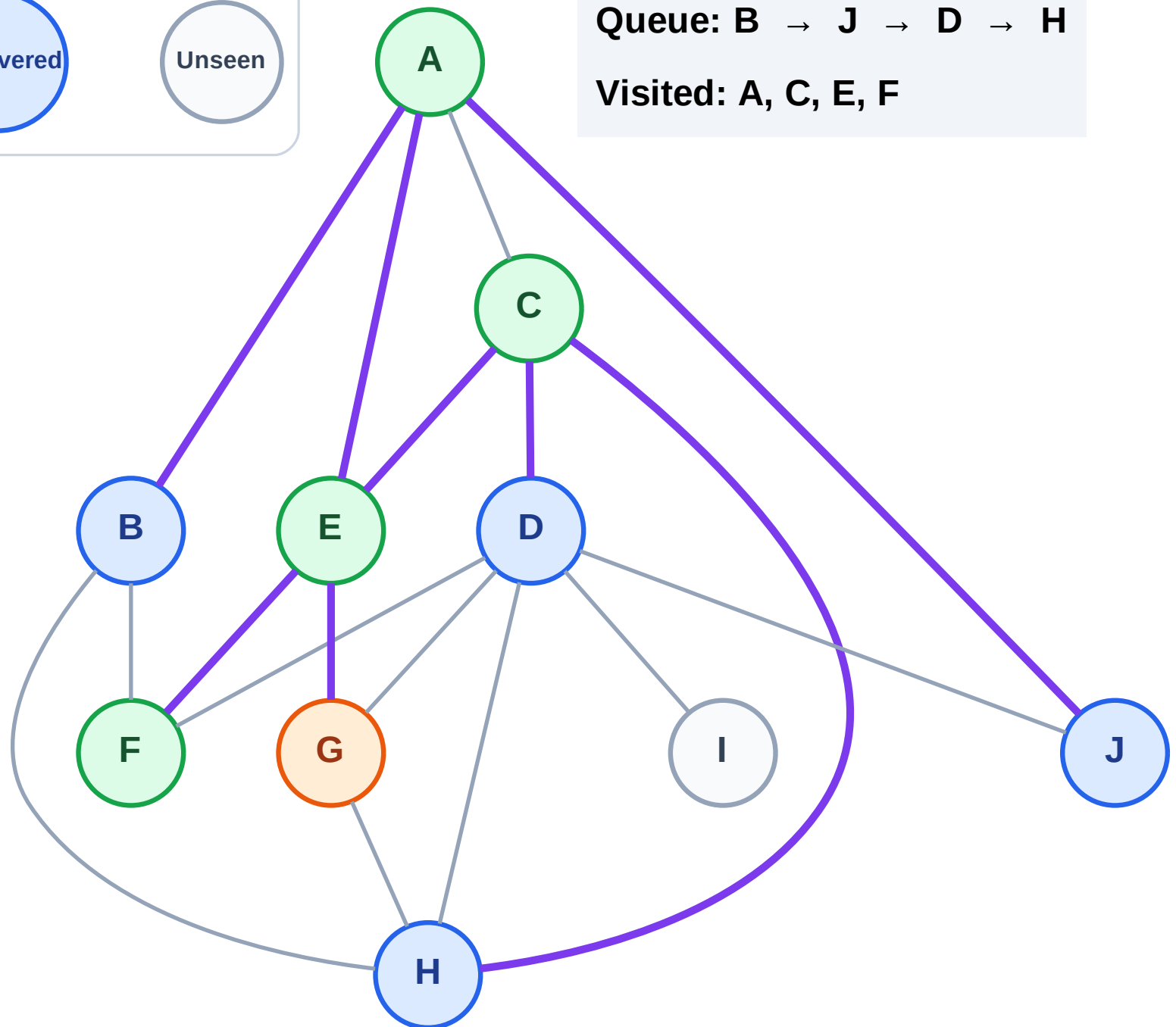
Processing

Discovered

Unseen

Queue: B → J → D → H

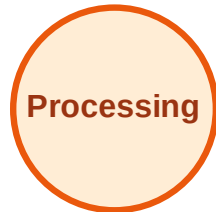
Visited: A, C, E, F



BFS Traversal

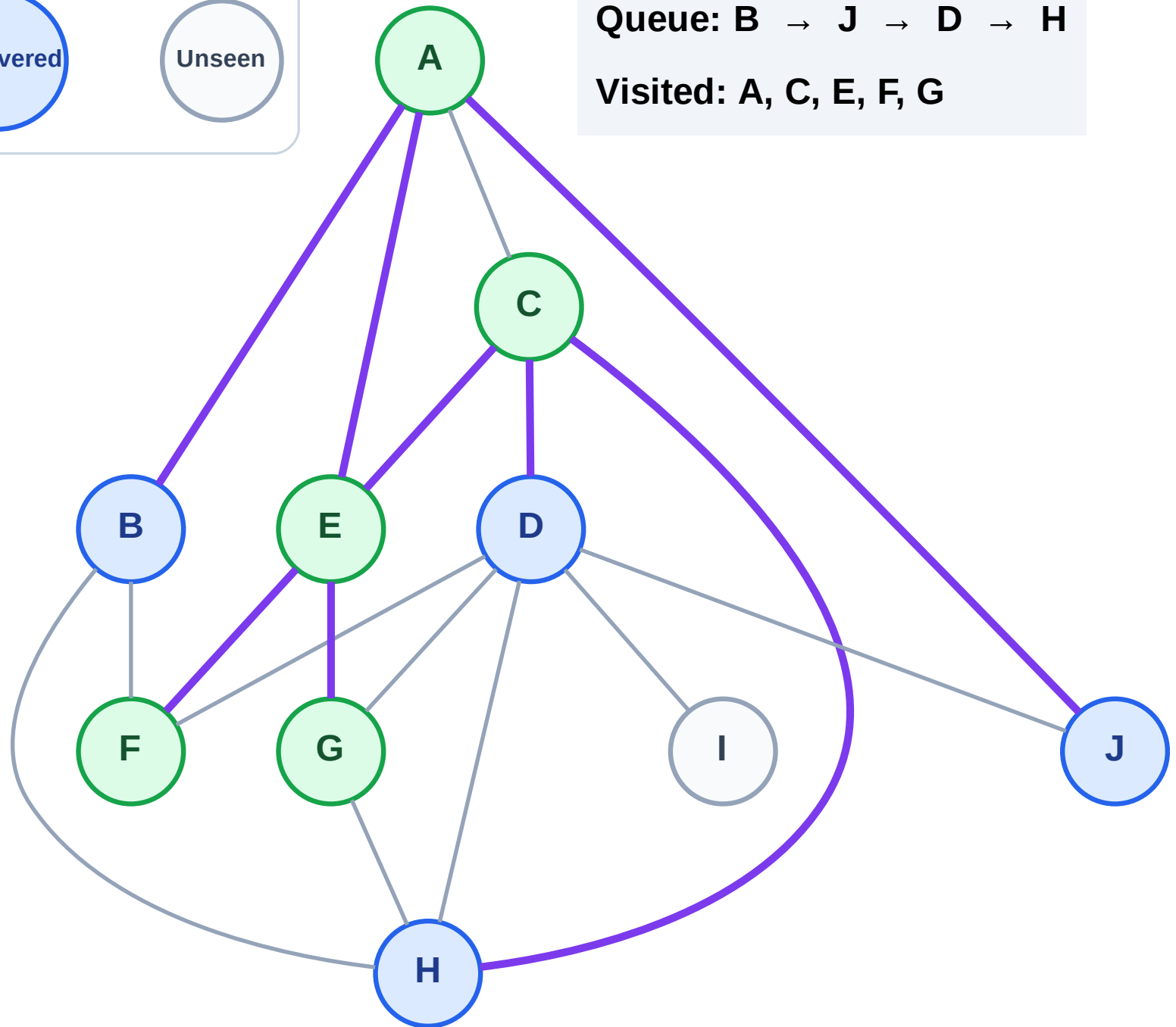
Step 11: No new neighbors from G

Legend



Queue: B → J → D → H

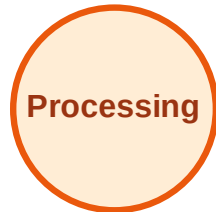
Visited: A, C, E, F, G



BFS Traversal

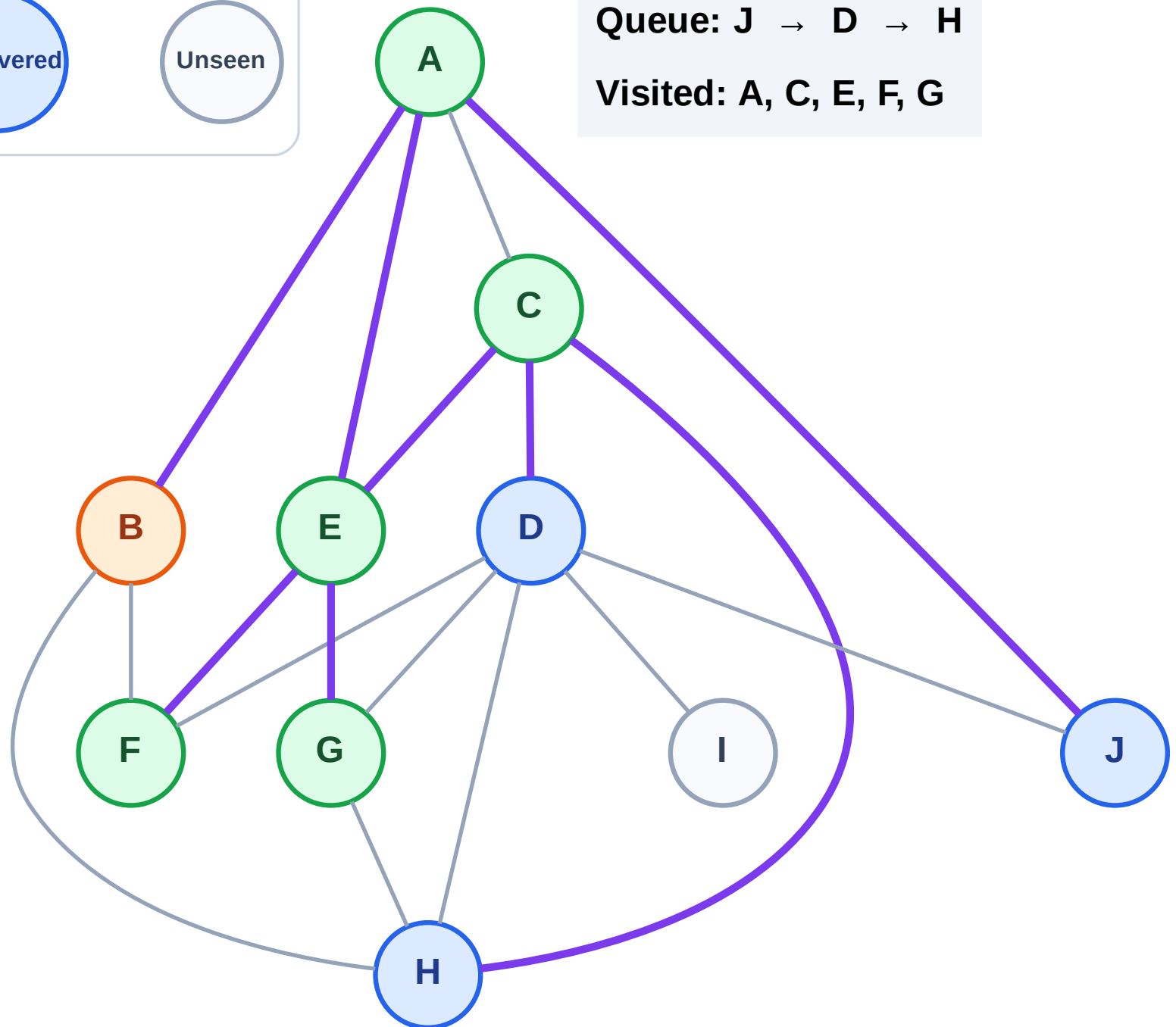
Step 12: Process node B

Legend



Queue: J → D → H

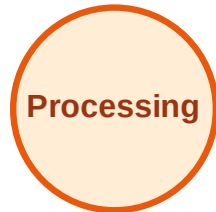
Visited: A, C, E, F, G



BFS Traversal

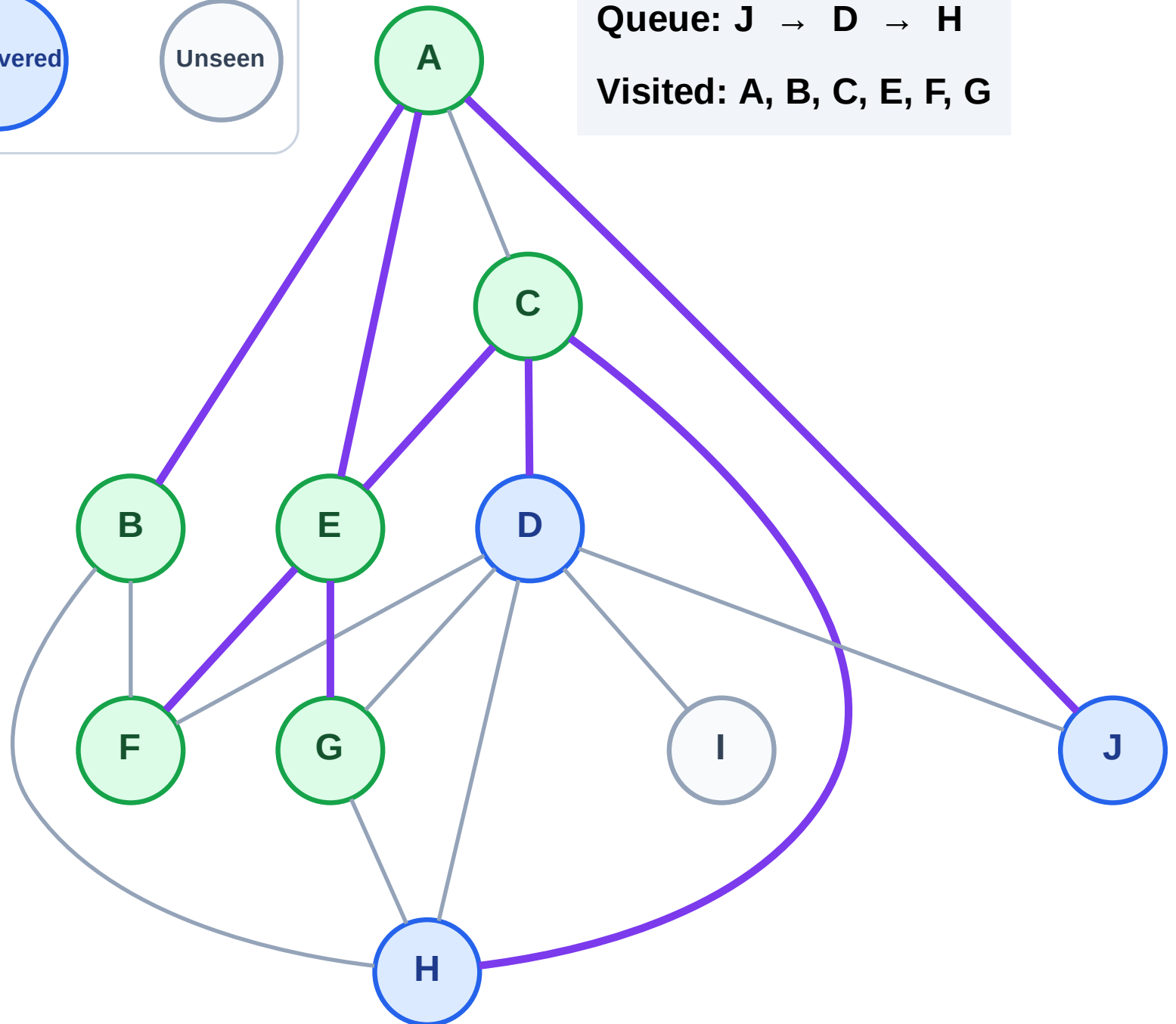
Step 13: No new neighbors from B

Legend



Queue: J → D → H

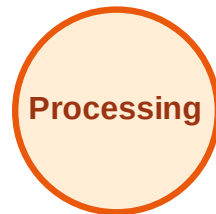
Visited: A, B, C, E, F, G



BFS Traversal

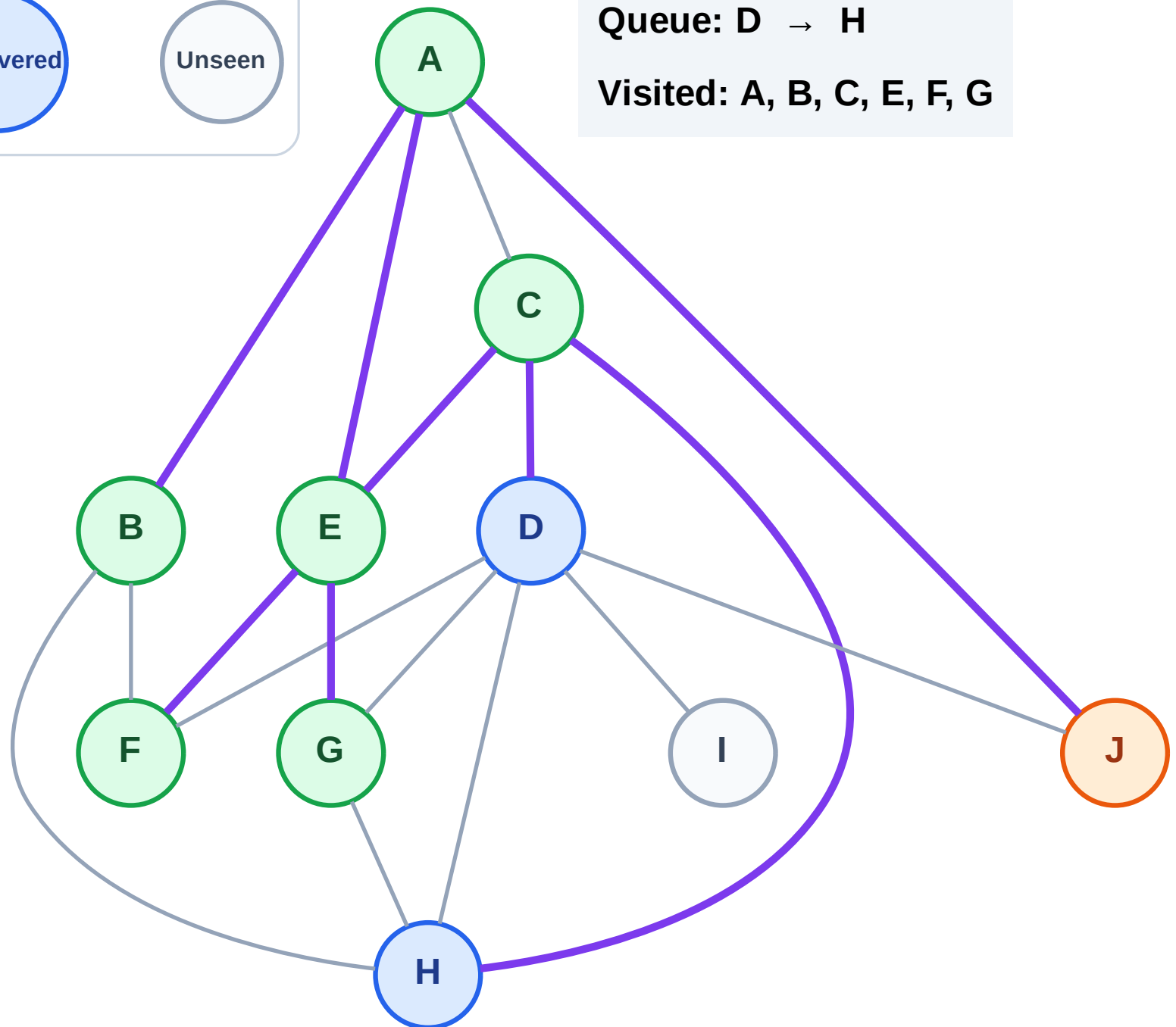
Step 14: Process node J

Legend



Queue: D → H

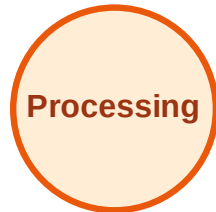
Visited: A, B, C, E, F, G



BFS Traversal

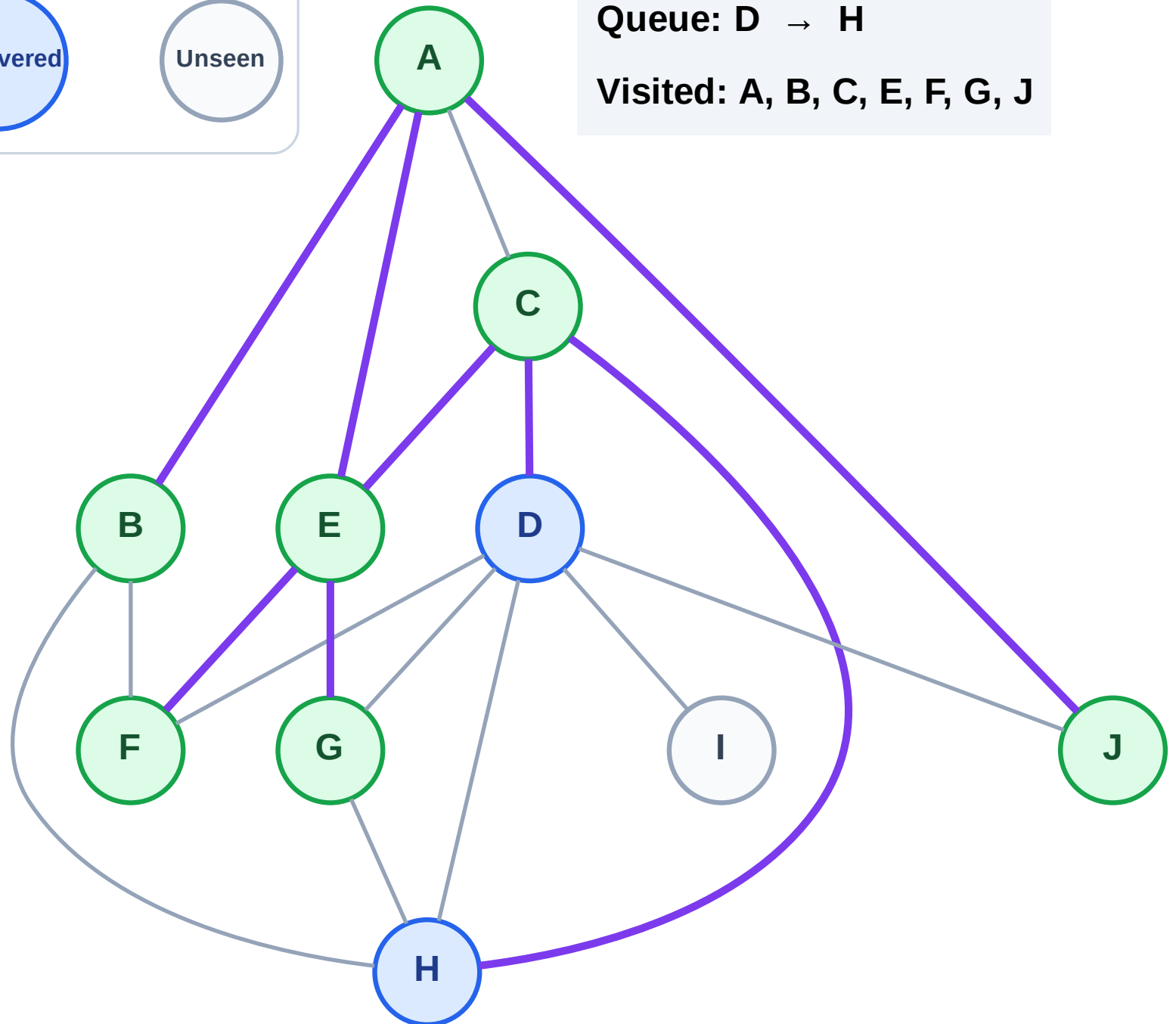
Step 15: No new neighbors from J

Legend



Queue: D → H

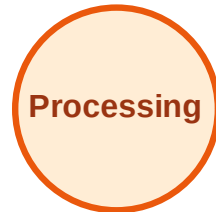
Visited: A, B, C, E, F, G, J



BFS Traversal

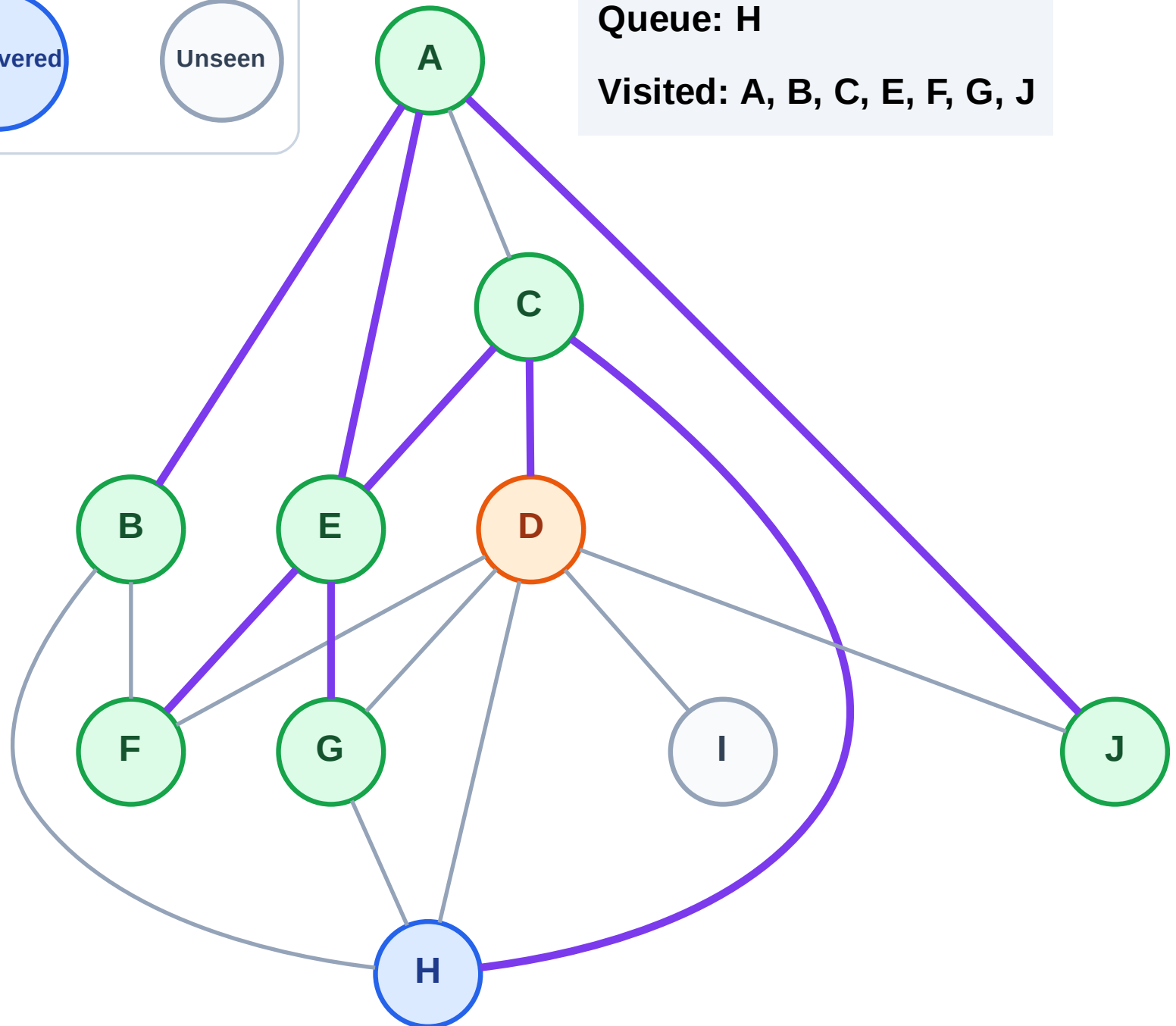
Step 16: Process node D

Legend



Queue: H

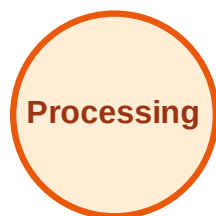
Visited: A, B, C, E, F, G, J



BFS Traversal

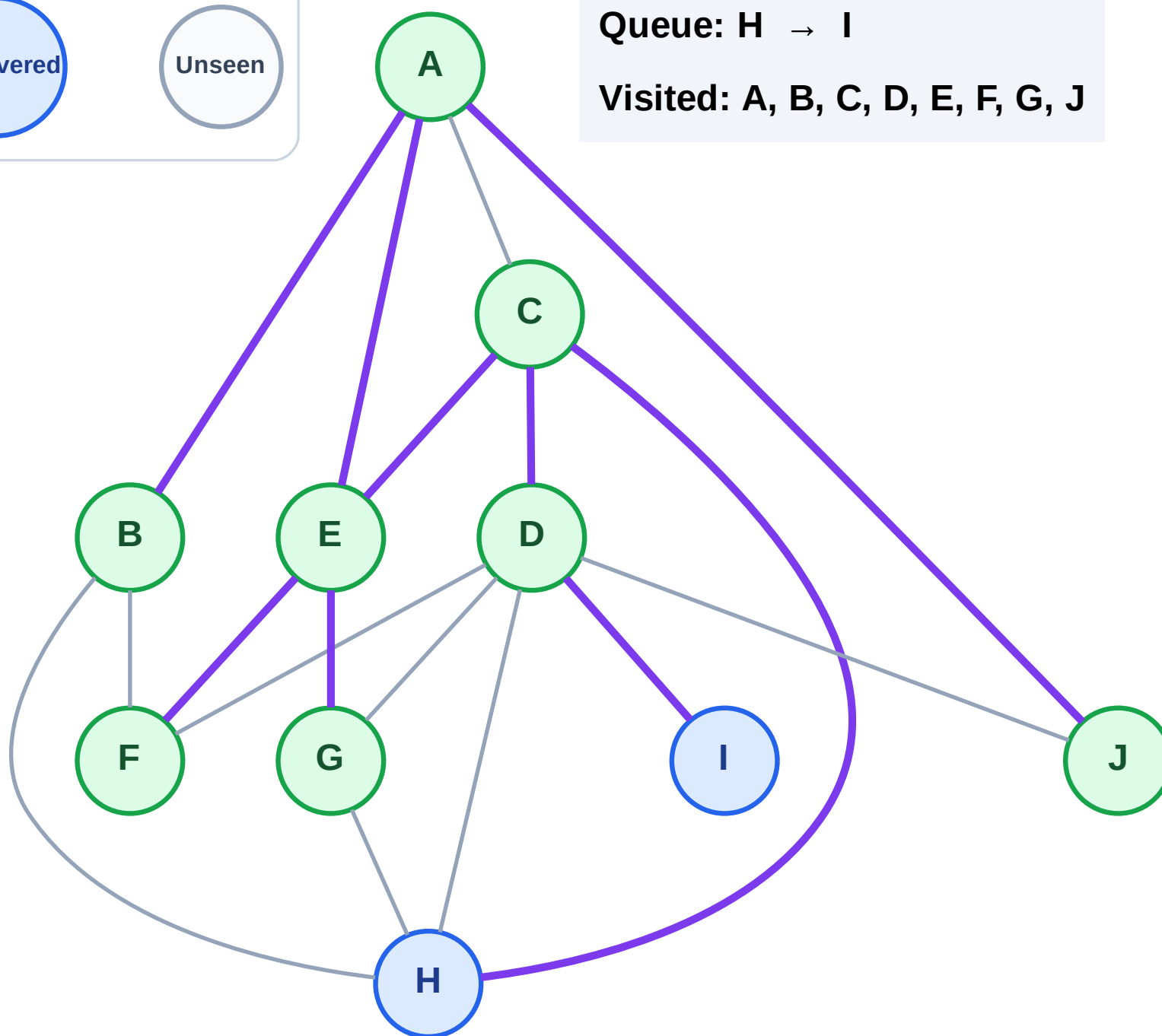
Step 17: Discover I from D

Legend



Queue: H → I

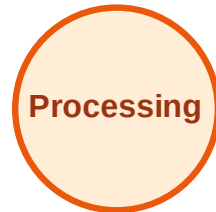
Visited: A, B, C, D, E, F, G, J



BFS Traversal

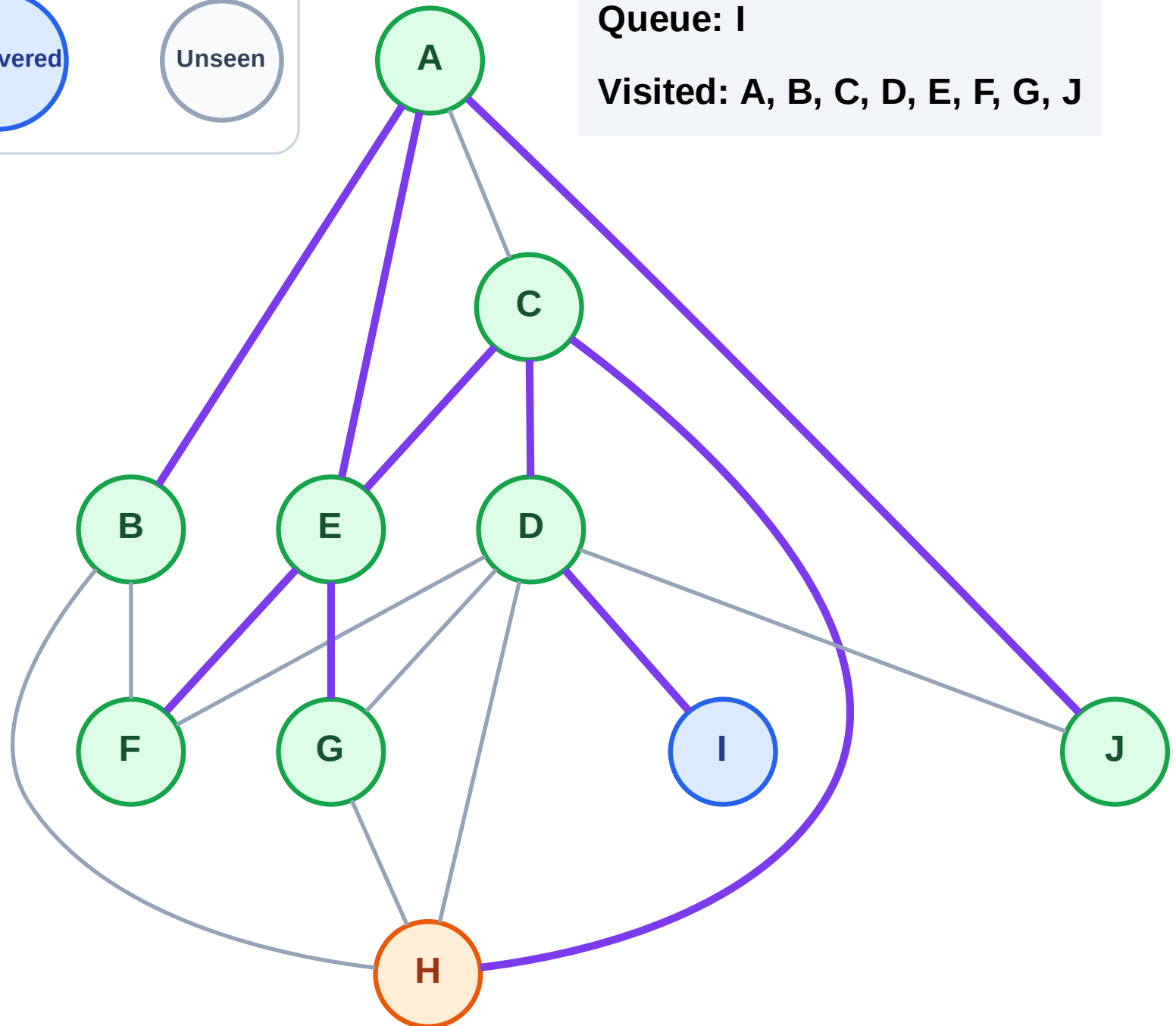
Step 18: Process node H

Legend



Queue: I

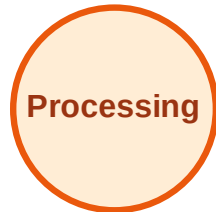
Visited: A, B, C, D, E, F, G, J



BFS Traversal

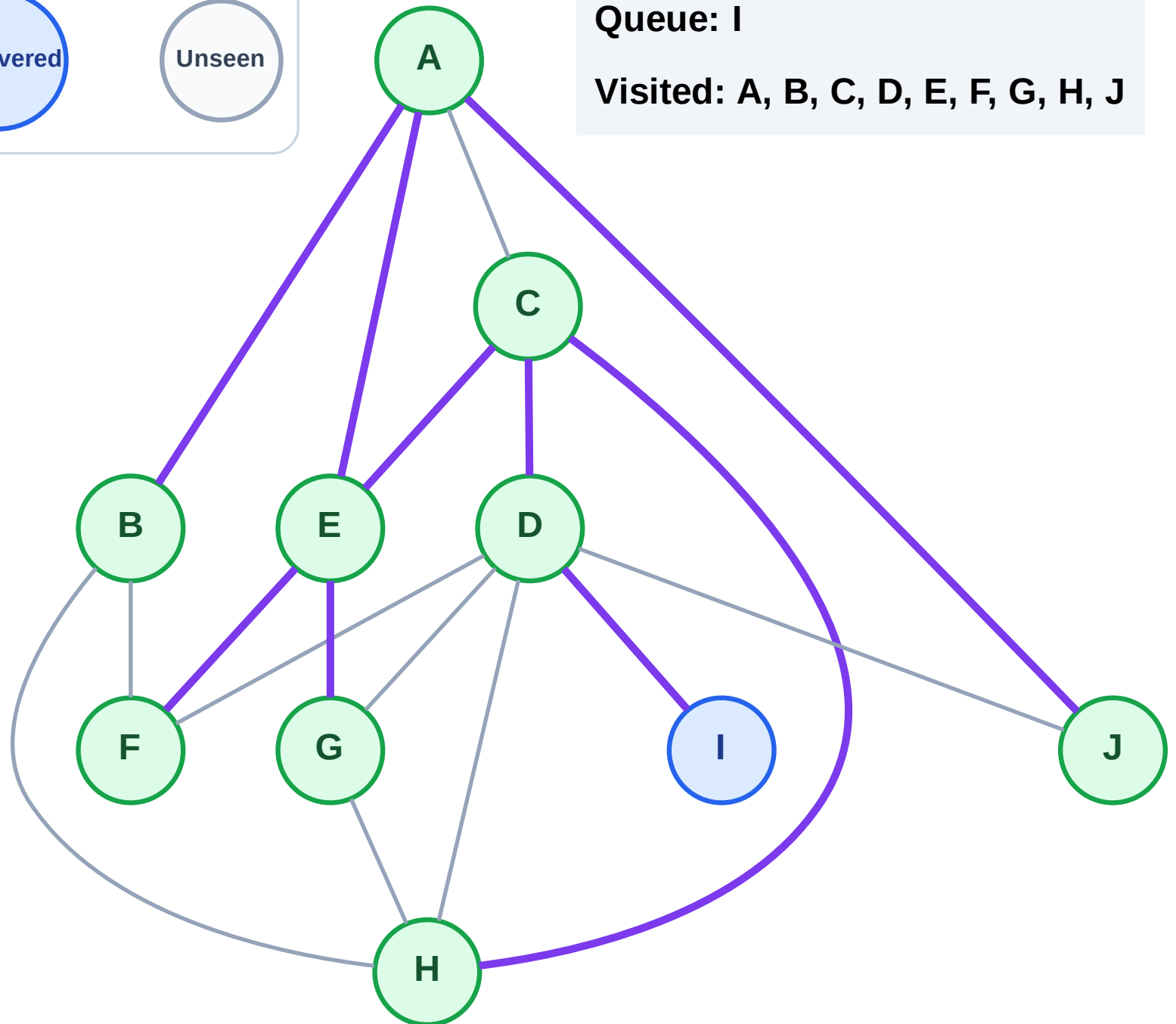
Step 19: No new neighbors from H

Legend



Queue: I

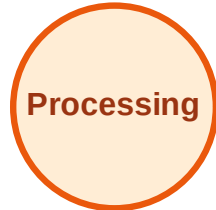
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

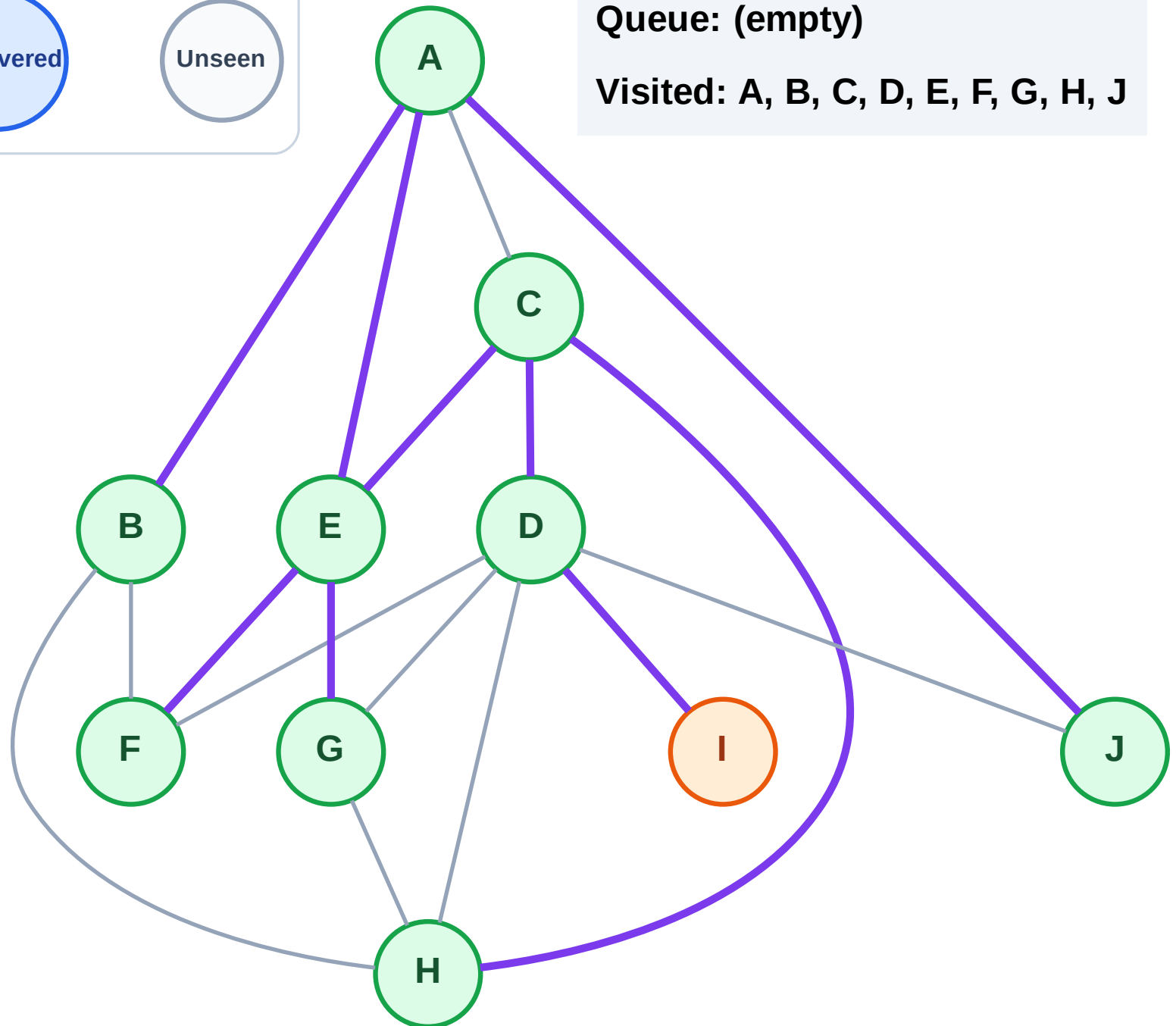
Step 20: Process node I

Legend



Queue: (empty)

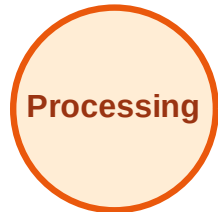
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

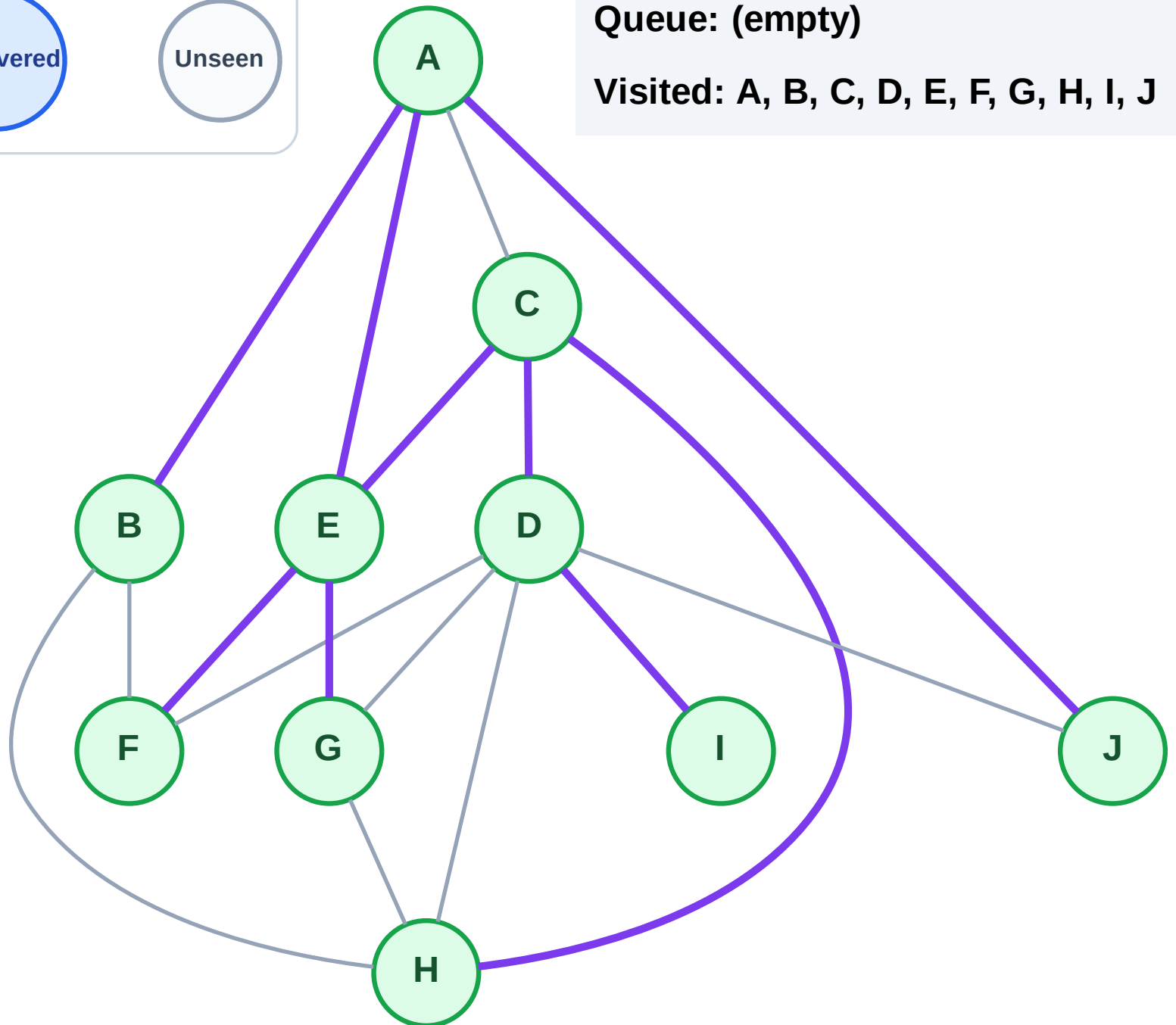
Step 21: No new neighbors from I

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

Step 1: Start at E

Legend

Complete

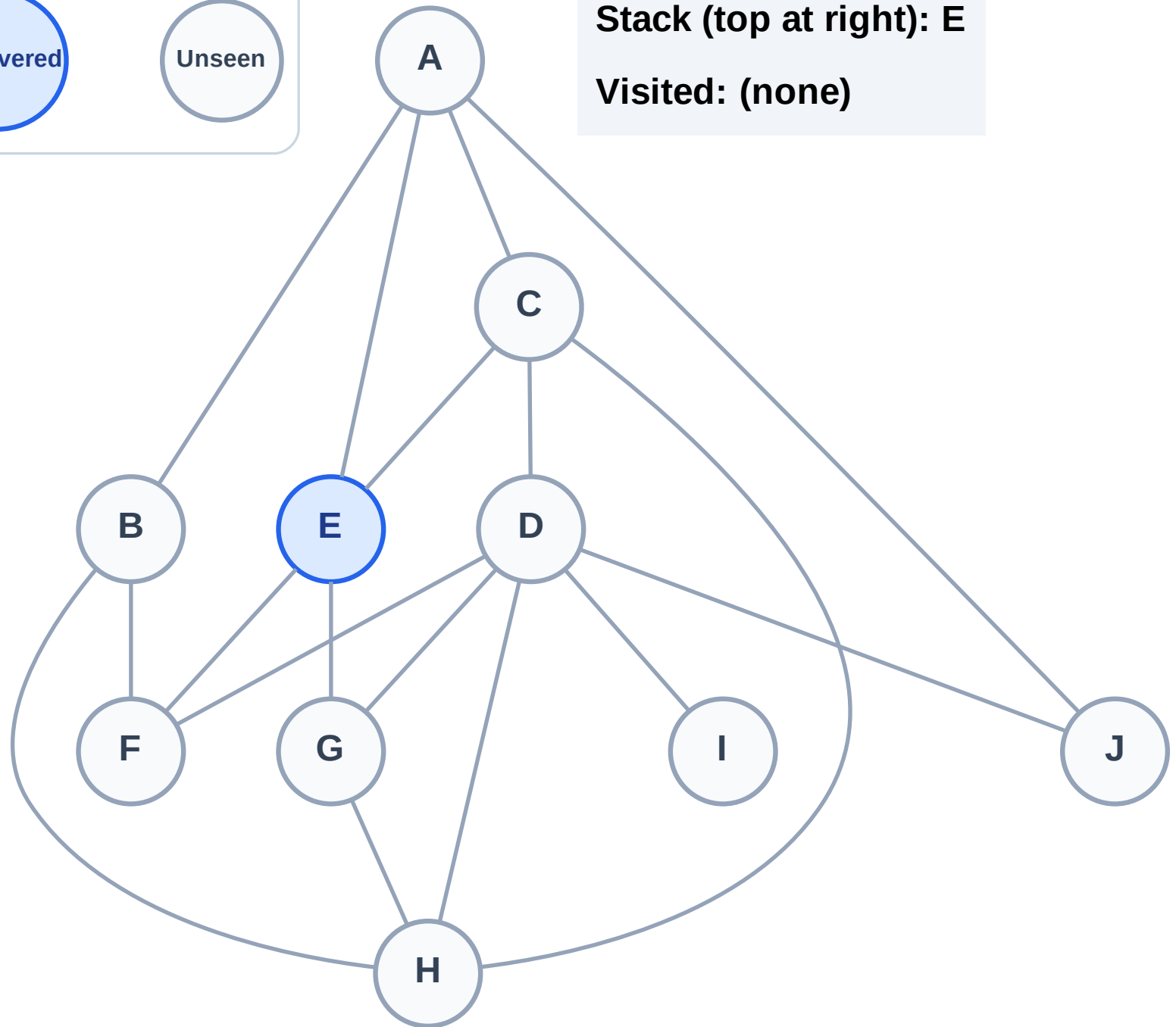
Processing

Discovered

Unseen

Stack (top at right): E

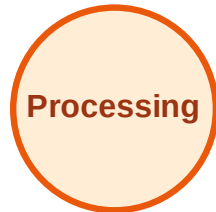
Visited: (none)



DFS Traversal

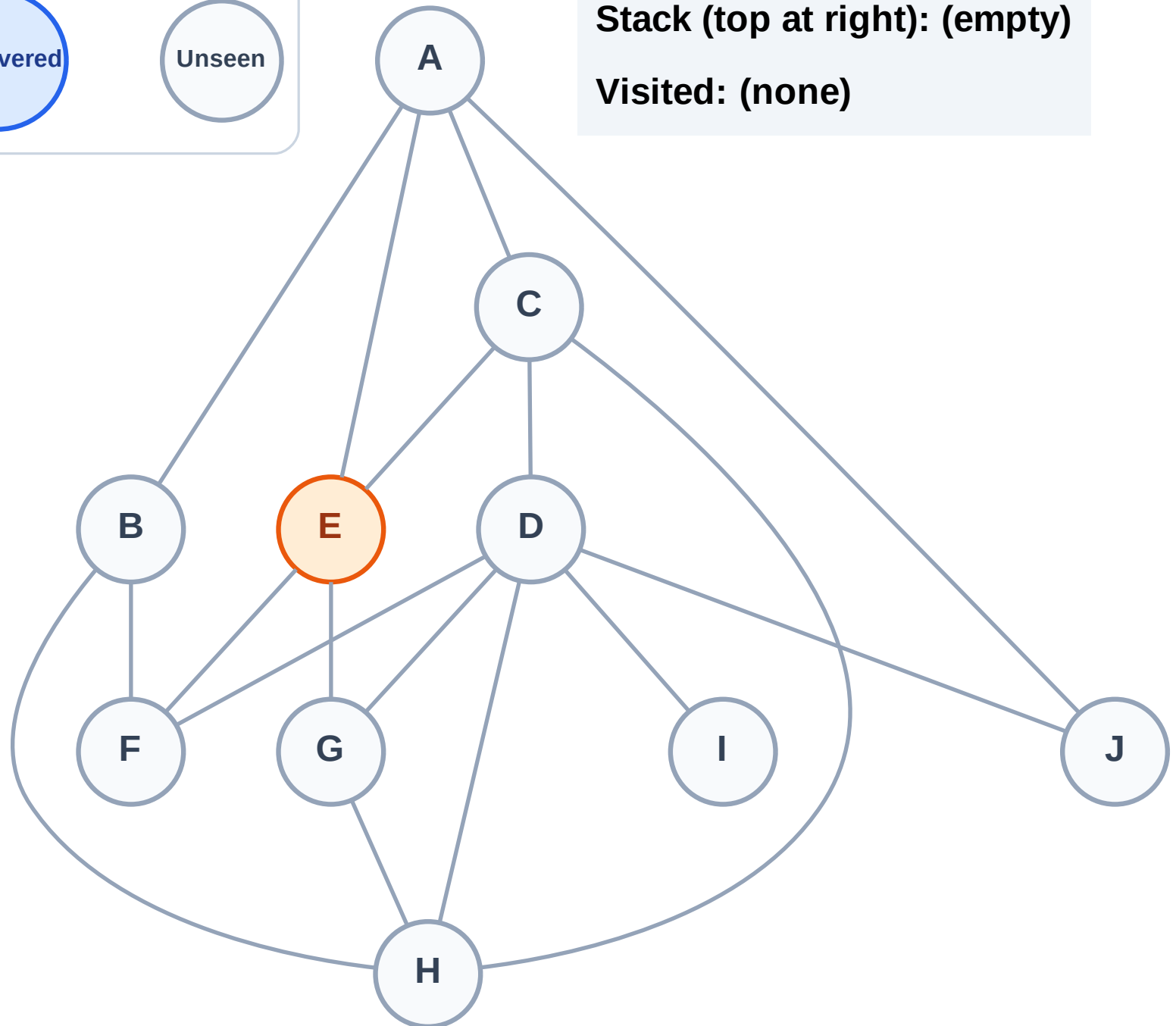
Step 2: Process node E

Legend



Stack (top at right): (empty)

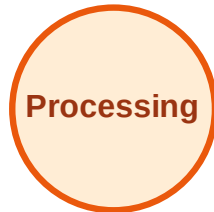
Visited: (none)



DFS Traversal

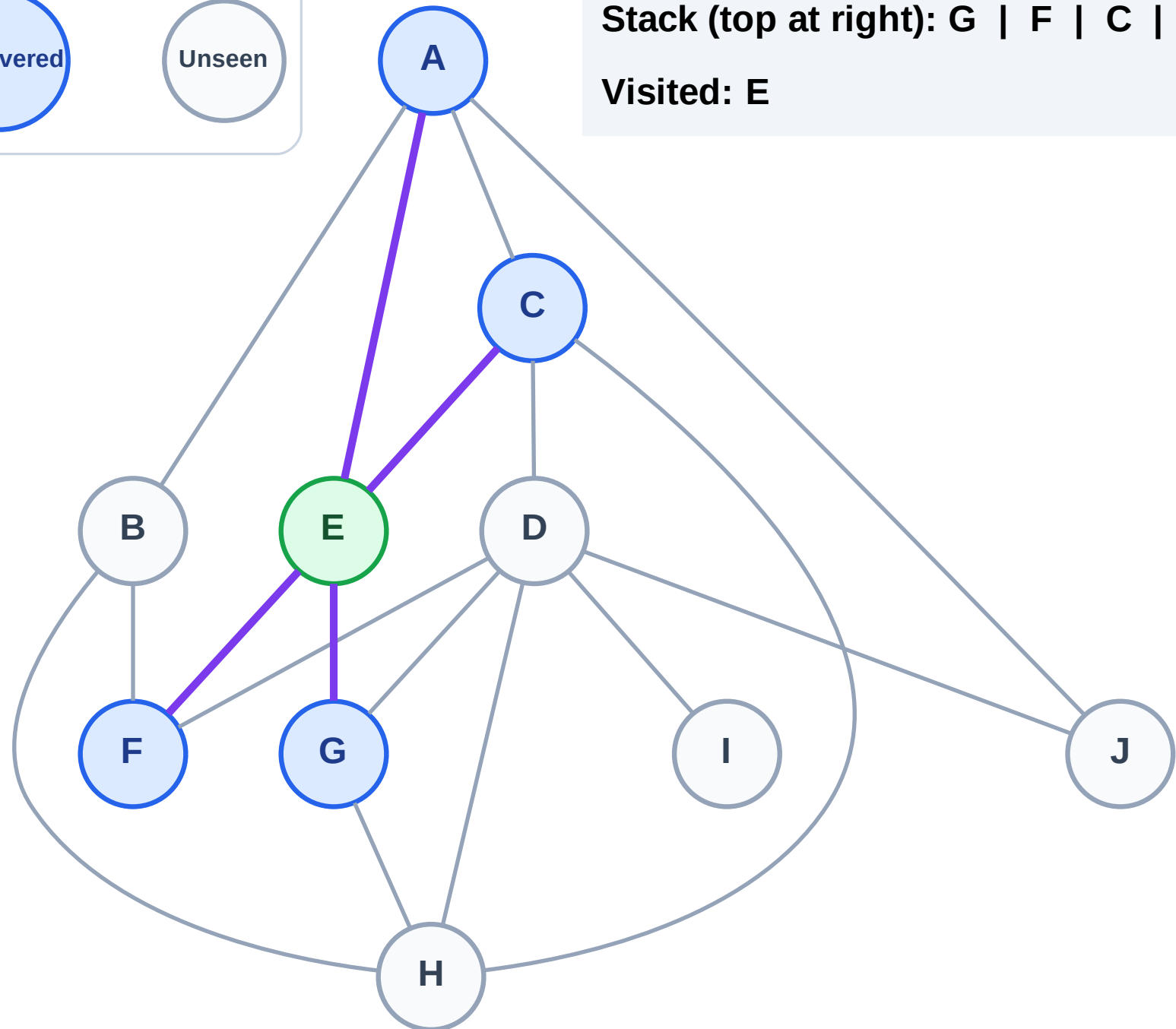
Step 3: Discover G, F, C, A from E

Legend



Stack (top at right): G | F | C | A

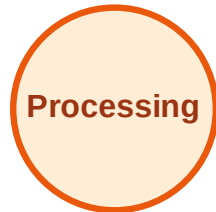
Visited: E



DFS Traversal

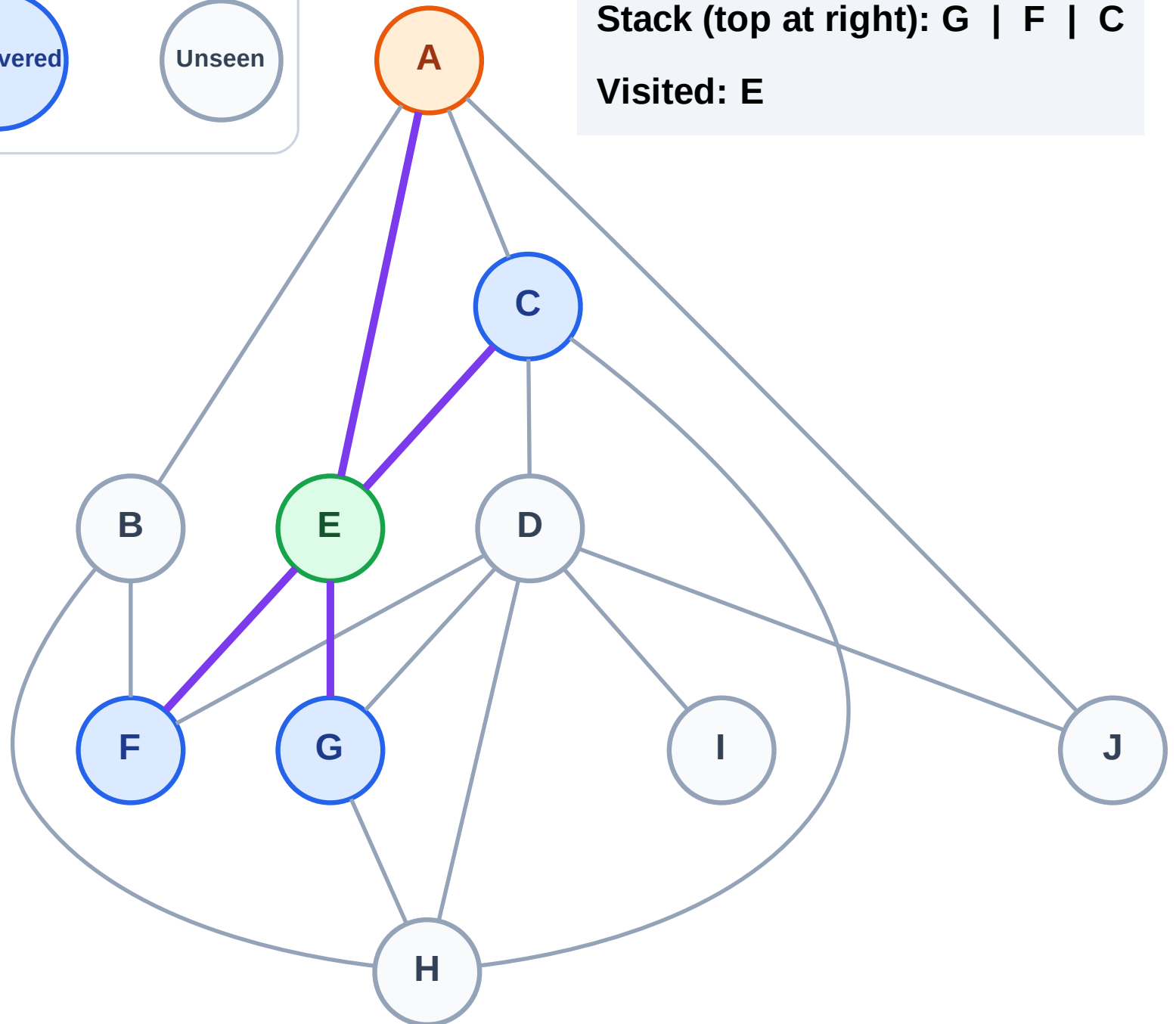
Step 4: Process node A

Legend



Stack (top at right): G | F | C

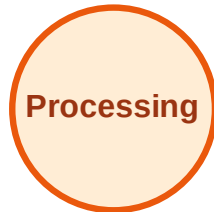
Visited: E



DFS Traversal

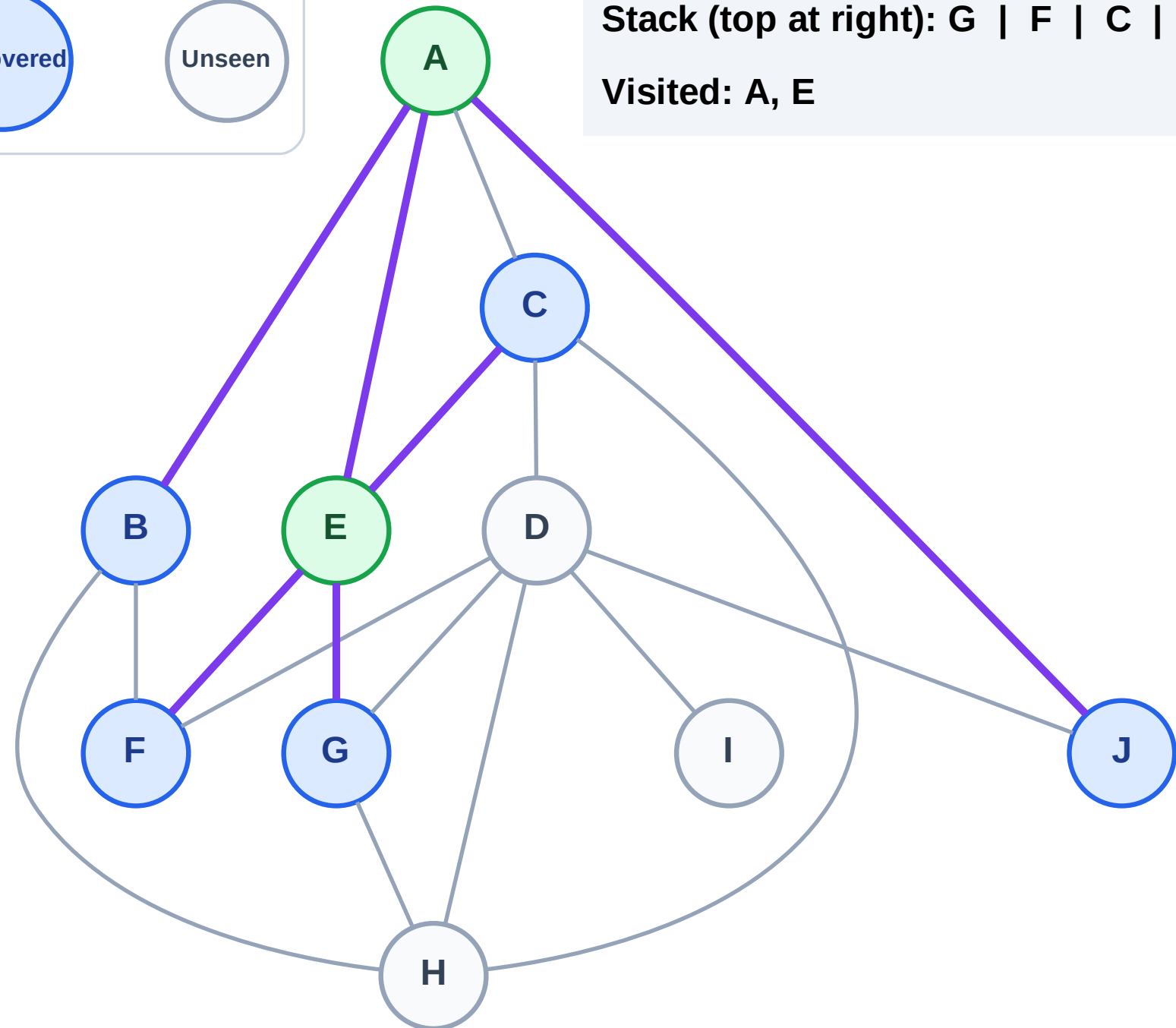
Step 5: Discover J, B from A

Legend



Stack (top at right): G | F | C | J | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

Complete

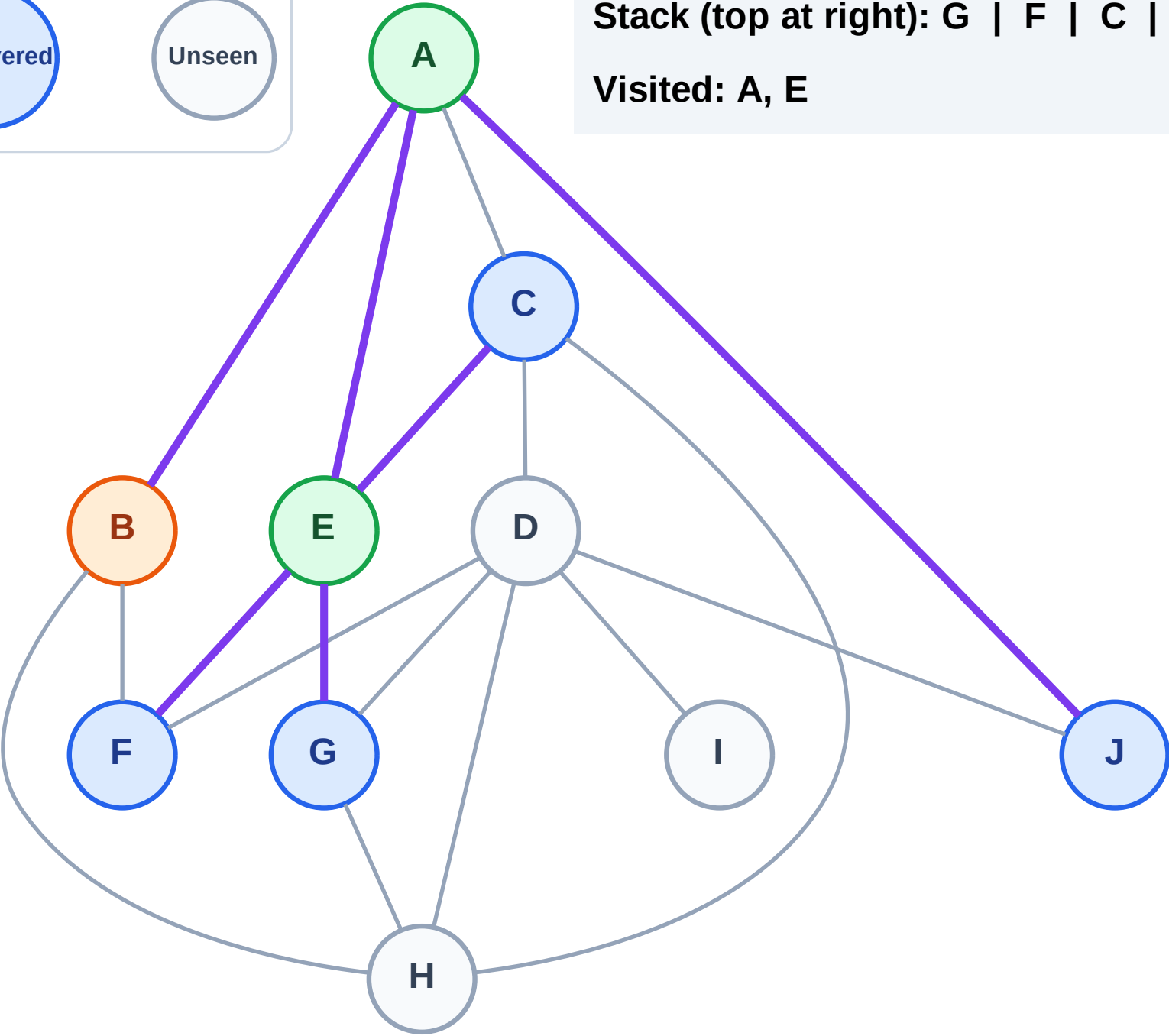
Processing

Discovered

Unseen

Stack (top at right): G | F | C | J

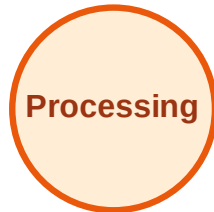
Visited: A, E



DFS Traversal

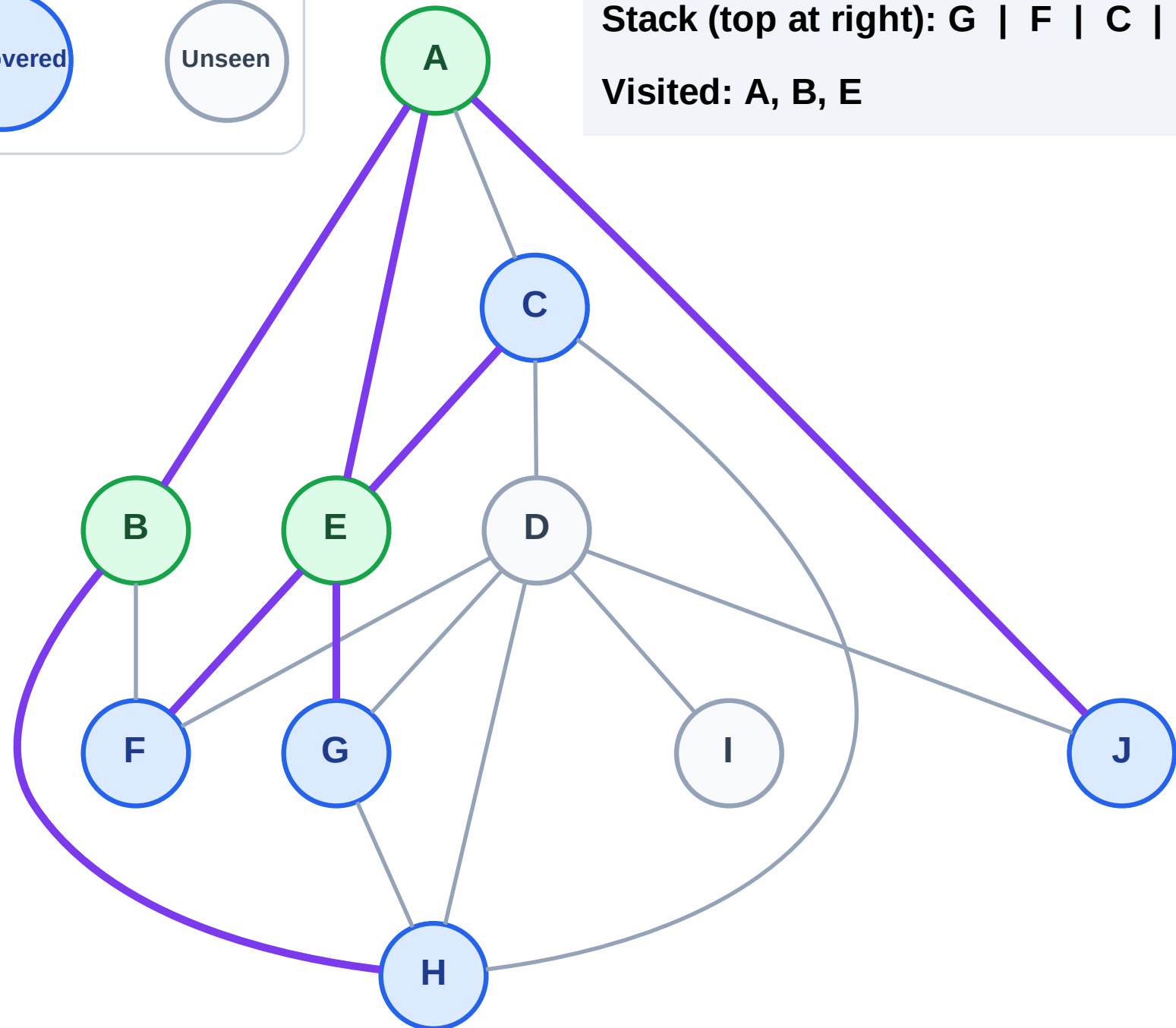
Step 7: Discover H from B

Legend

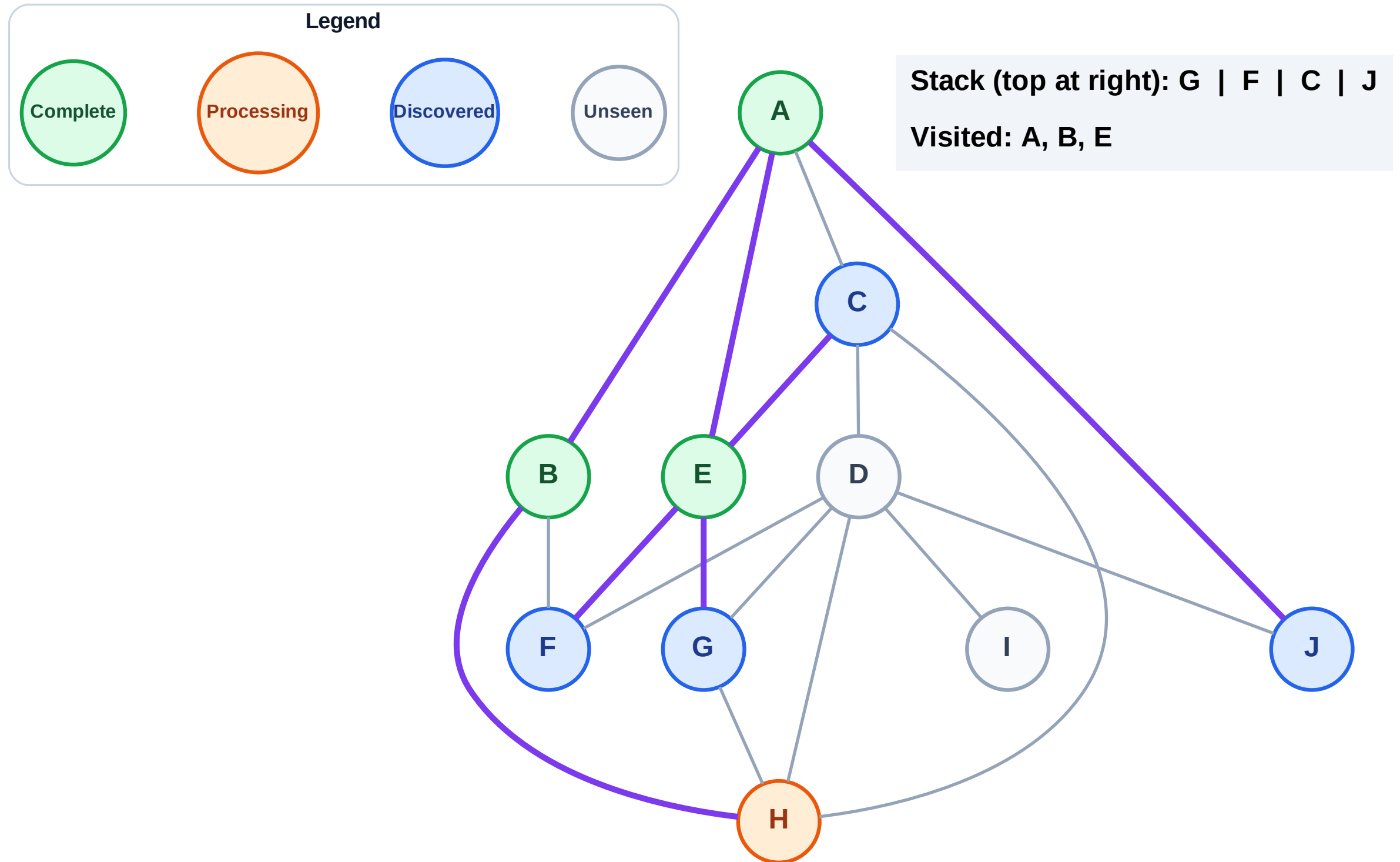


Stack (top at right): G | F | C | J | H

Visited: A, B, E

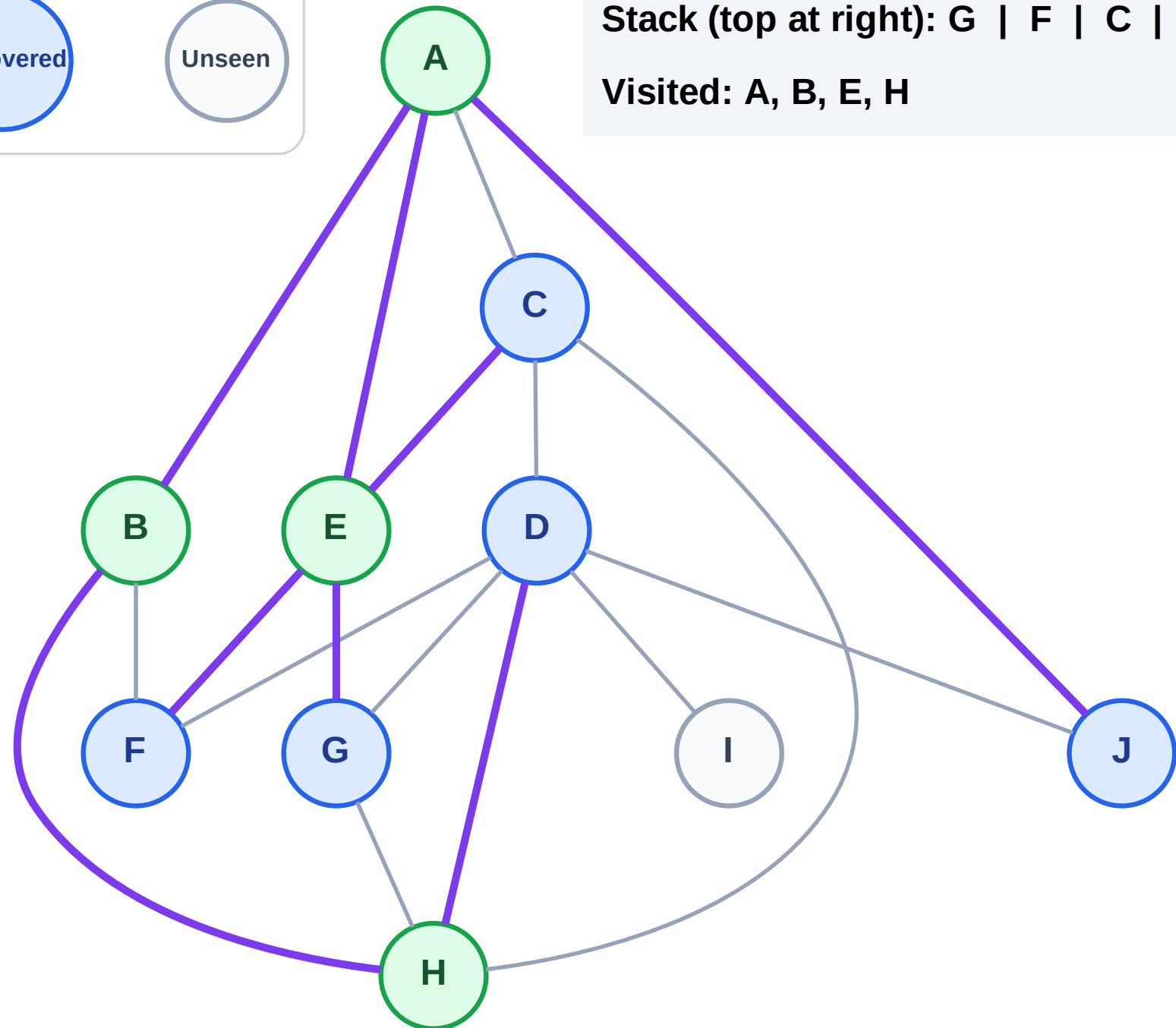
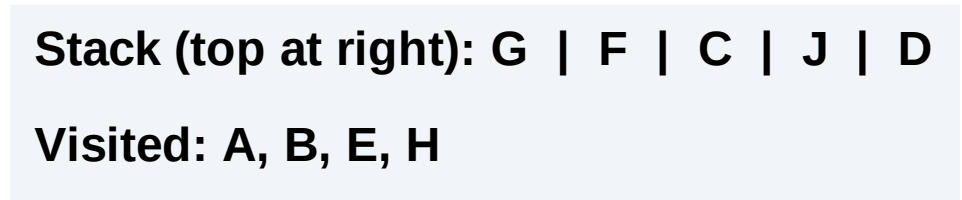


Step 8: Process node H



Step 9: Discover D from H

Step 9: Discover D from H



DFS Traversal

Step 10: Process node D

Legend

Complete

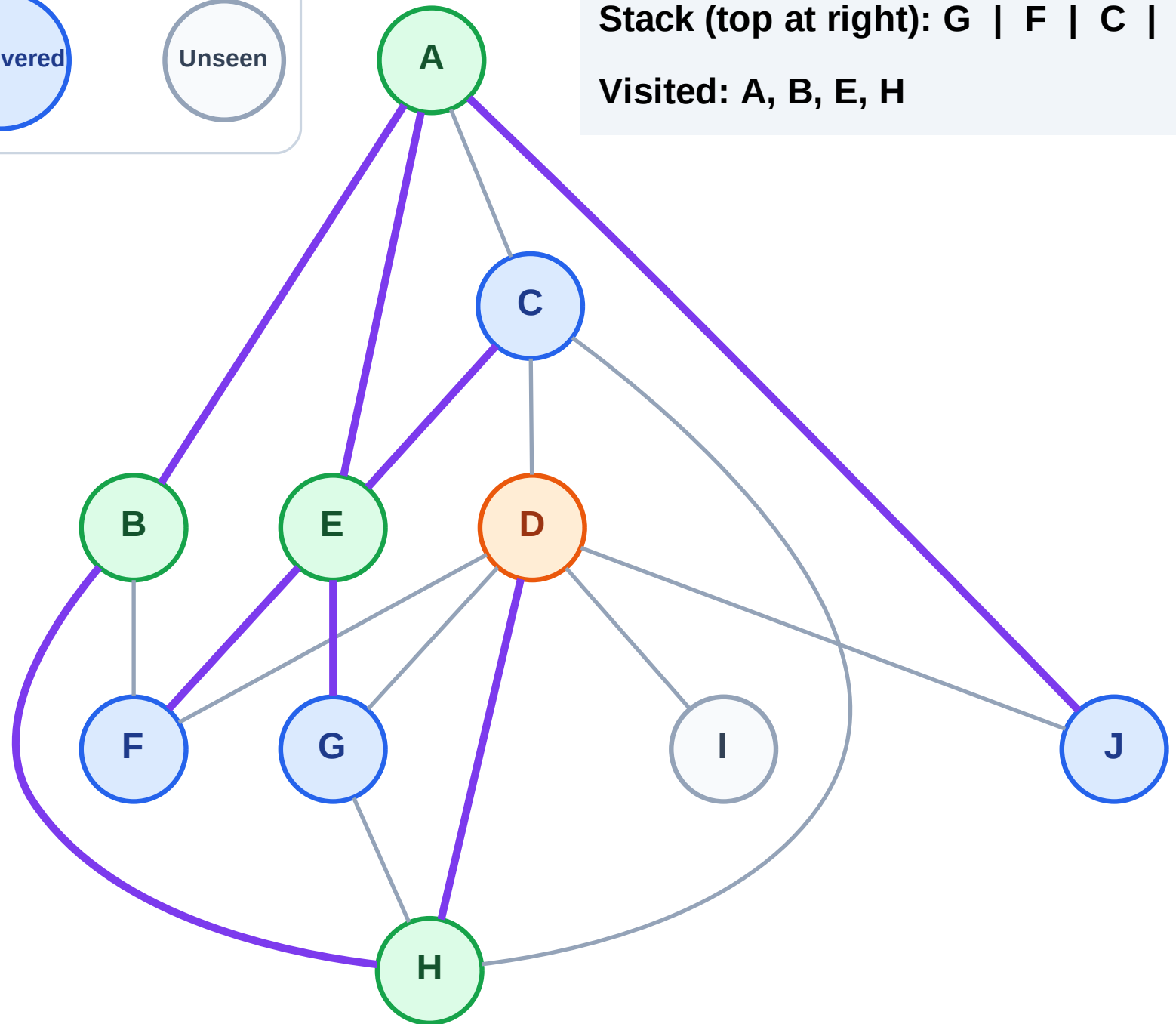
Processing

Discovered

Unseen

Stack (top at right): G | F | C | J

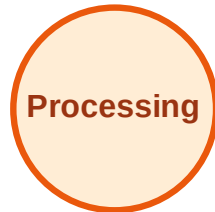
Visited: A, B, E, H



DFS Traversal

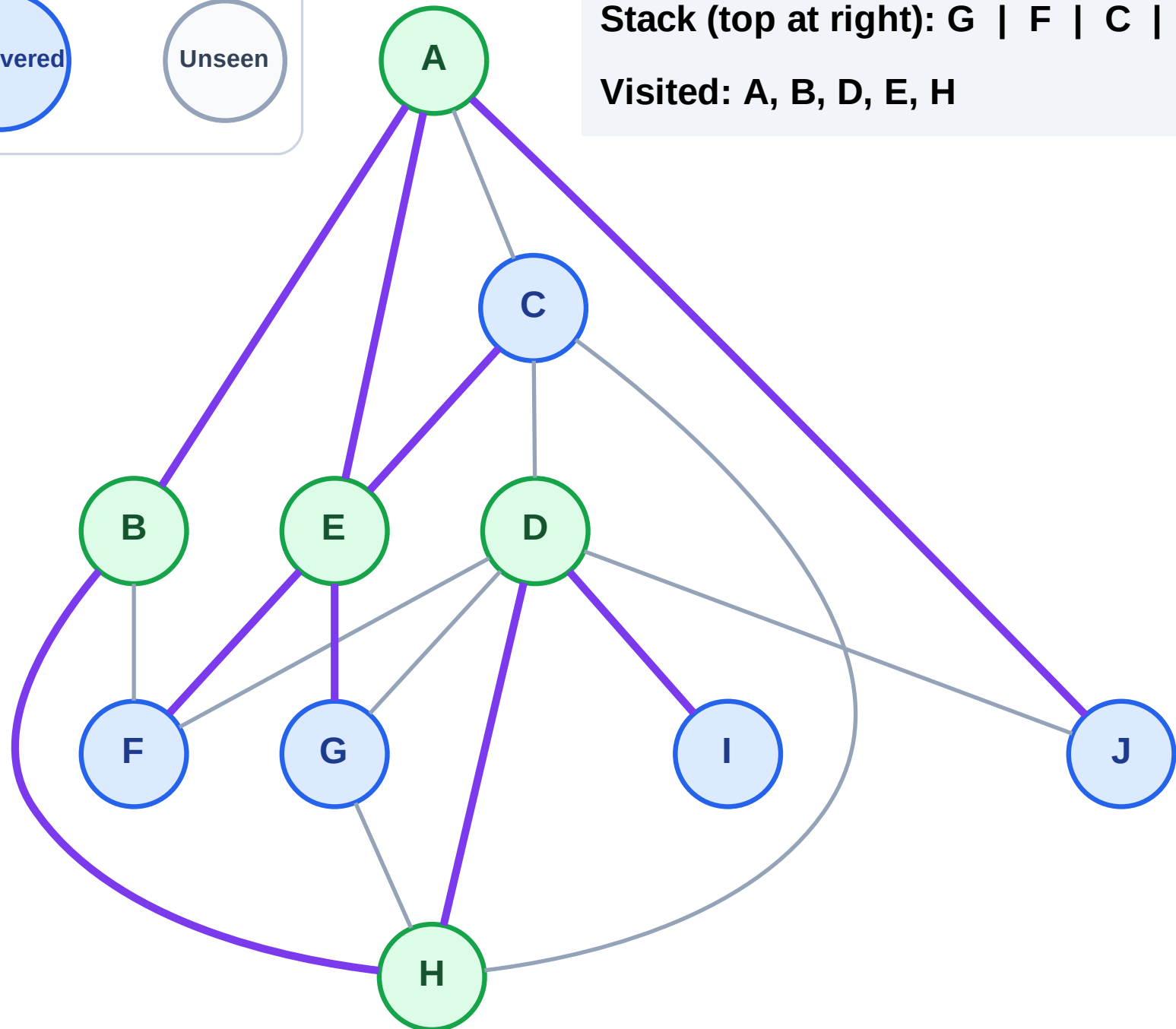
Step 11: Discover I from D

Legend



Stack (top at right): G | F | C | J | I

Visited: A, B, D, E, H



DFS Traversal

Step 12: Process node I

Legend

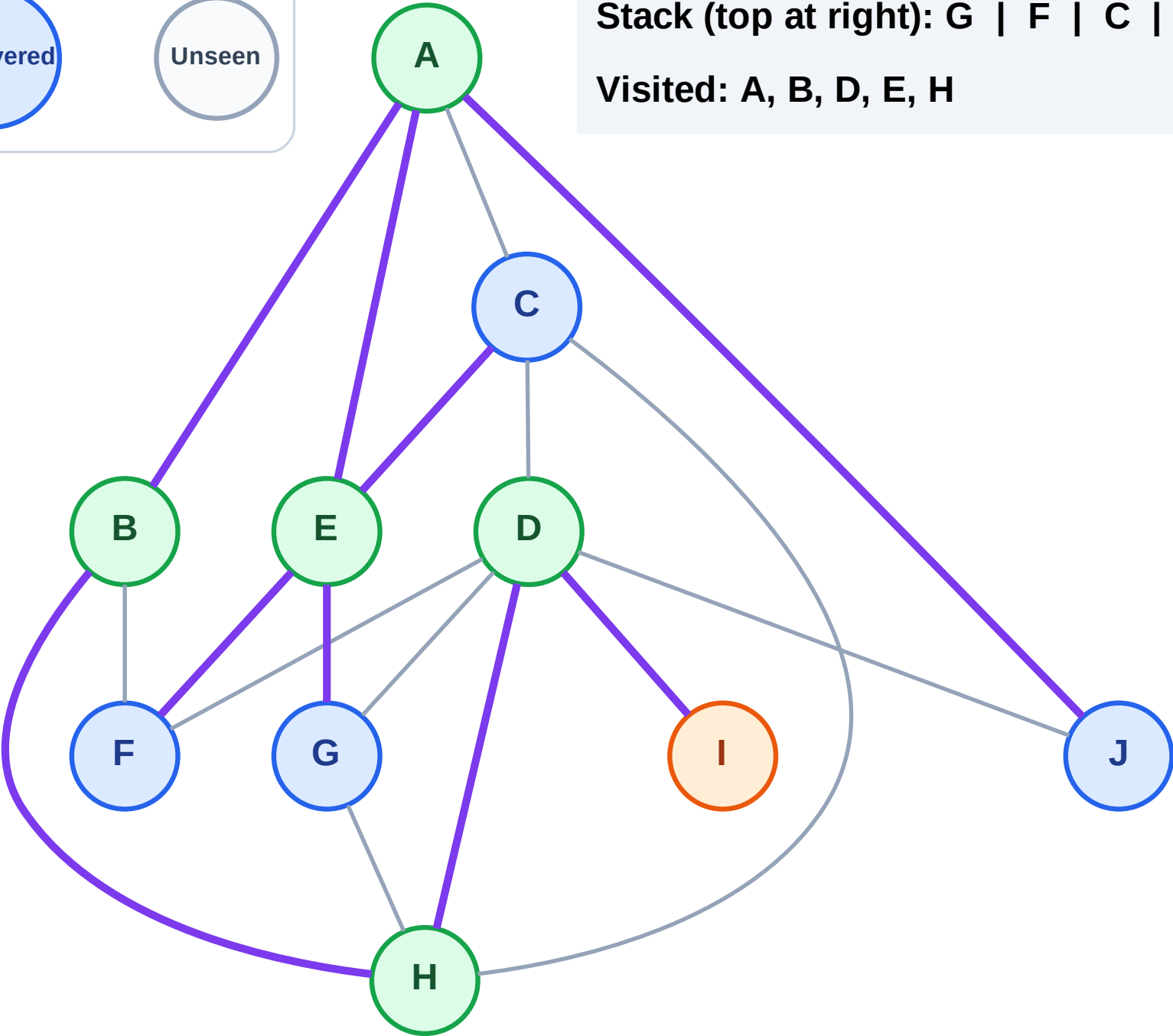
Complete

Processing

Discovered

Unseen

Stack (top at right): G | F | C | J
Visited: A, B, D, E, H



DFS Traversal

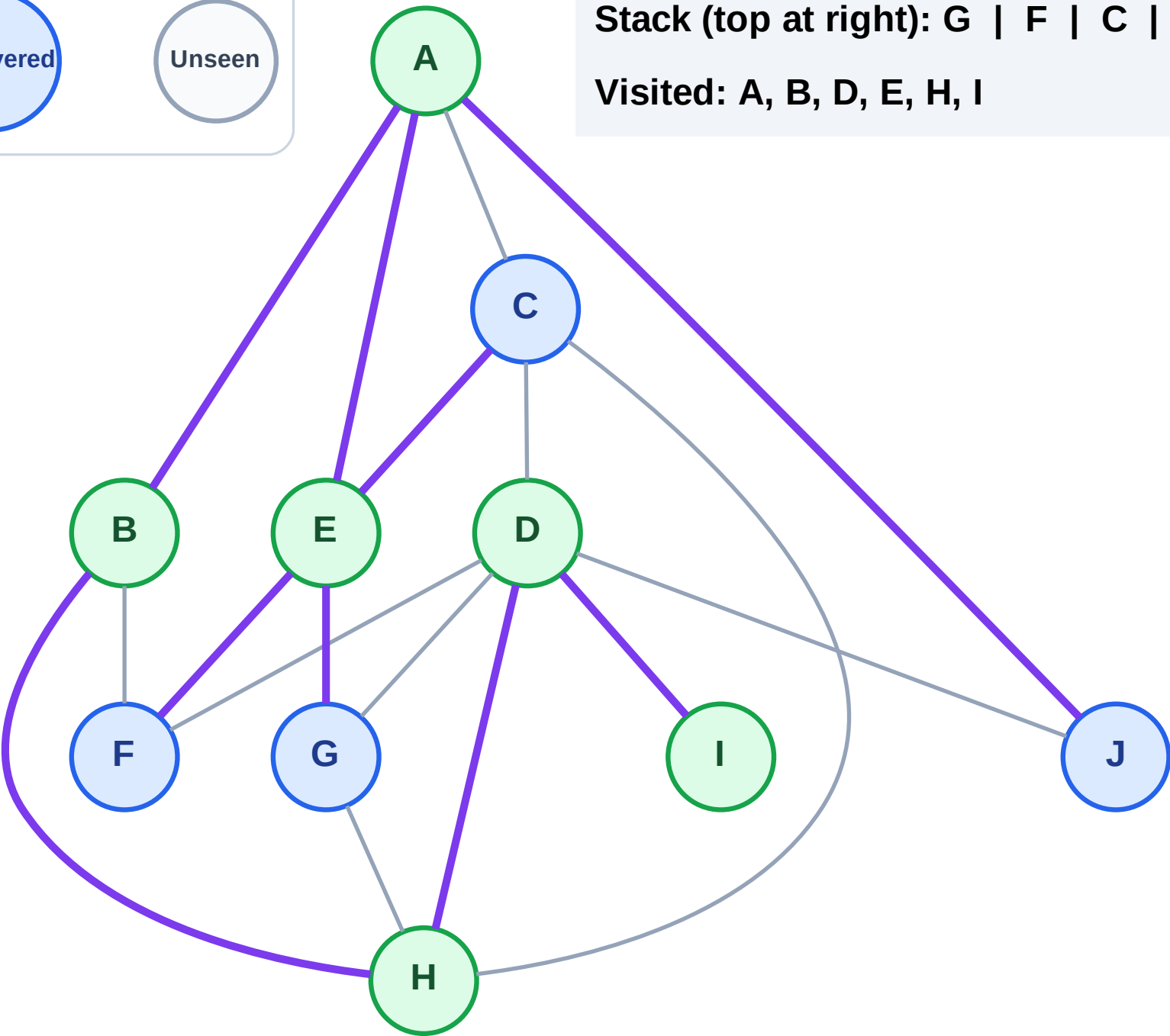
Step 13: No new neighbors from I

Legend



Stack (top at right): G | F | C | J

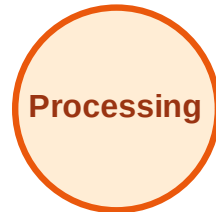
Visited: A, B, D, E, H, I



DFS Traversal

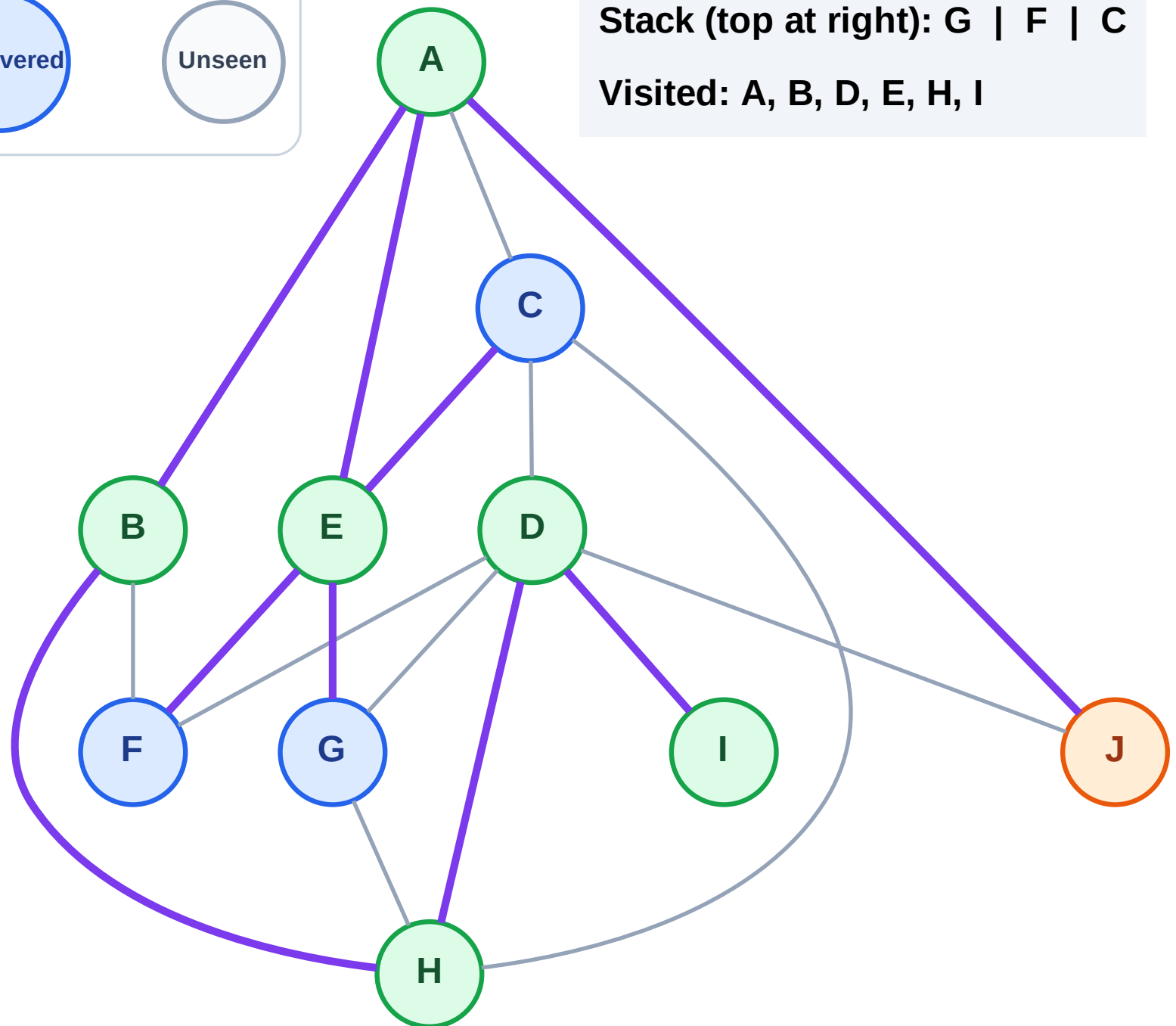
Step 14: Process node J

Legend



Stack (top at right): G | F | C

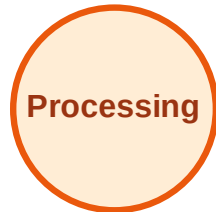
Visited: A, B, D, E, H, I



DFS Traversal

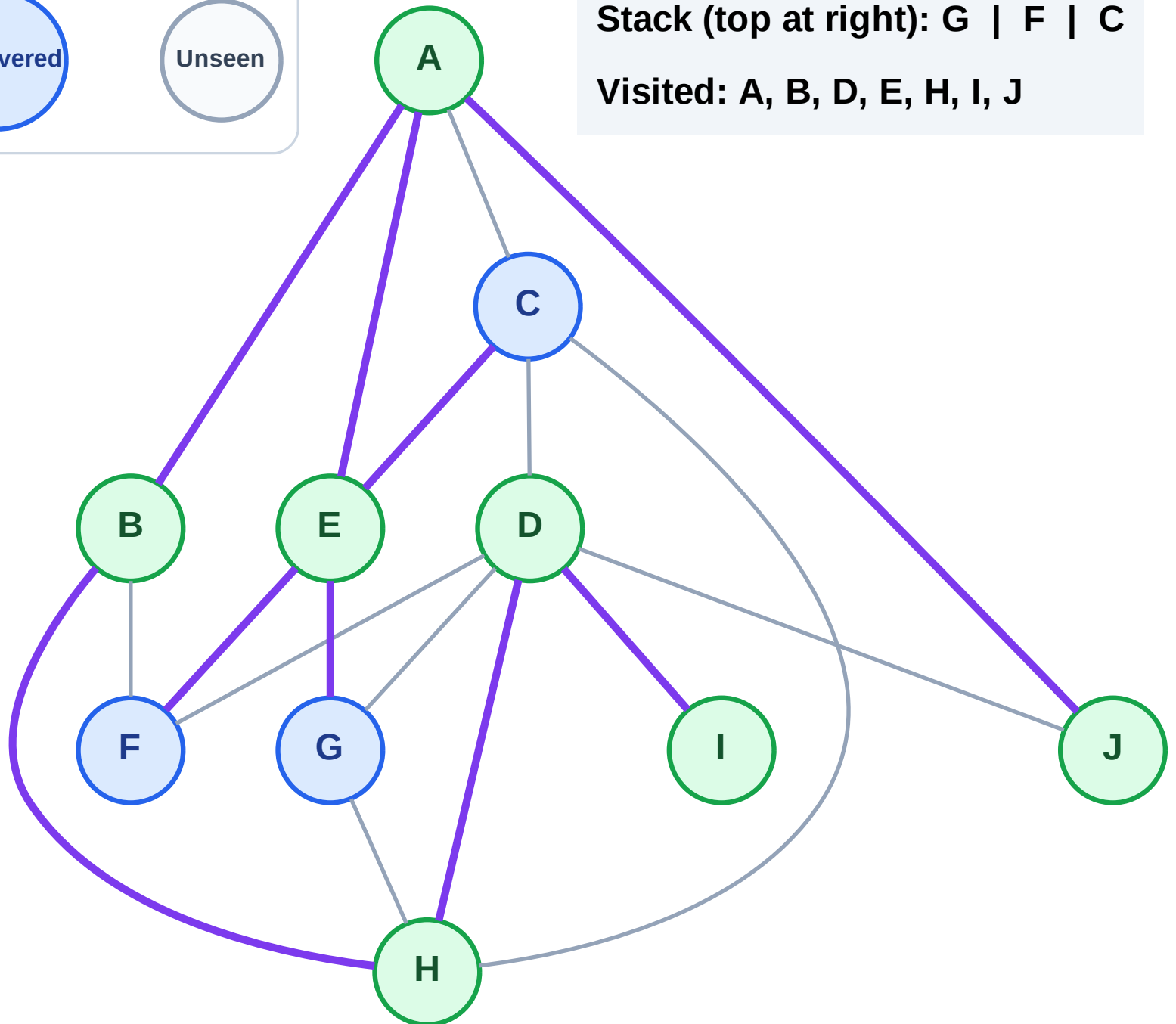
Step 15: No new neighbors from J

Legend



Stack (top at right): G | F | C

Visited: A, B, D, E, H, I, J



DFS Traversal

Step 16: Process node C

Legend

Complete

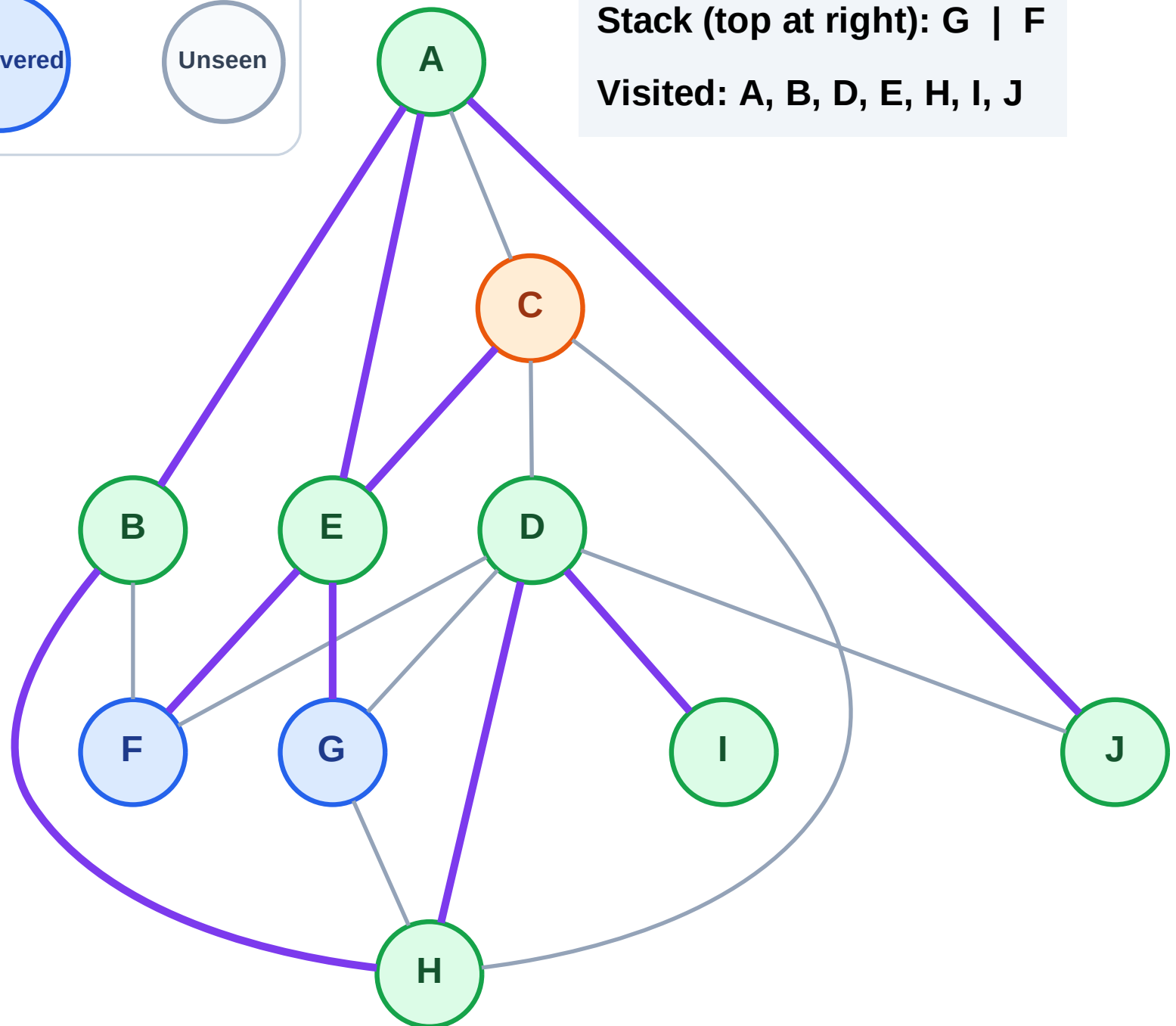
Processing

Discovered

Unseen

Stack (top at right): G | F

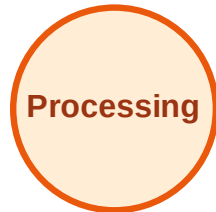
Visited: A, B, D, E, H, I, J



DFS Traversal

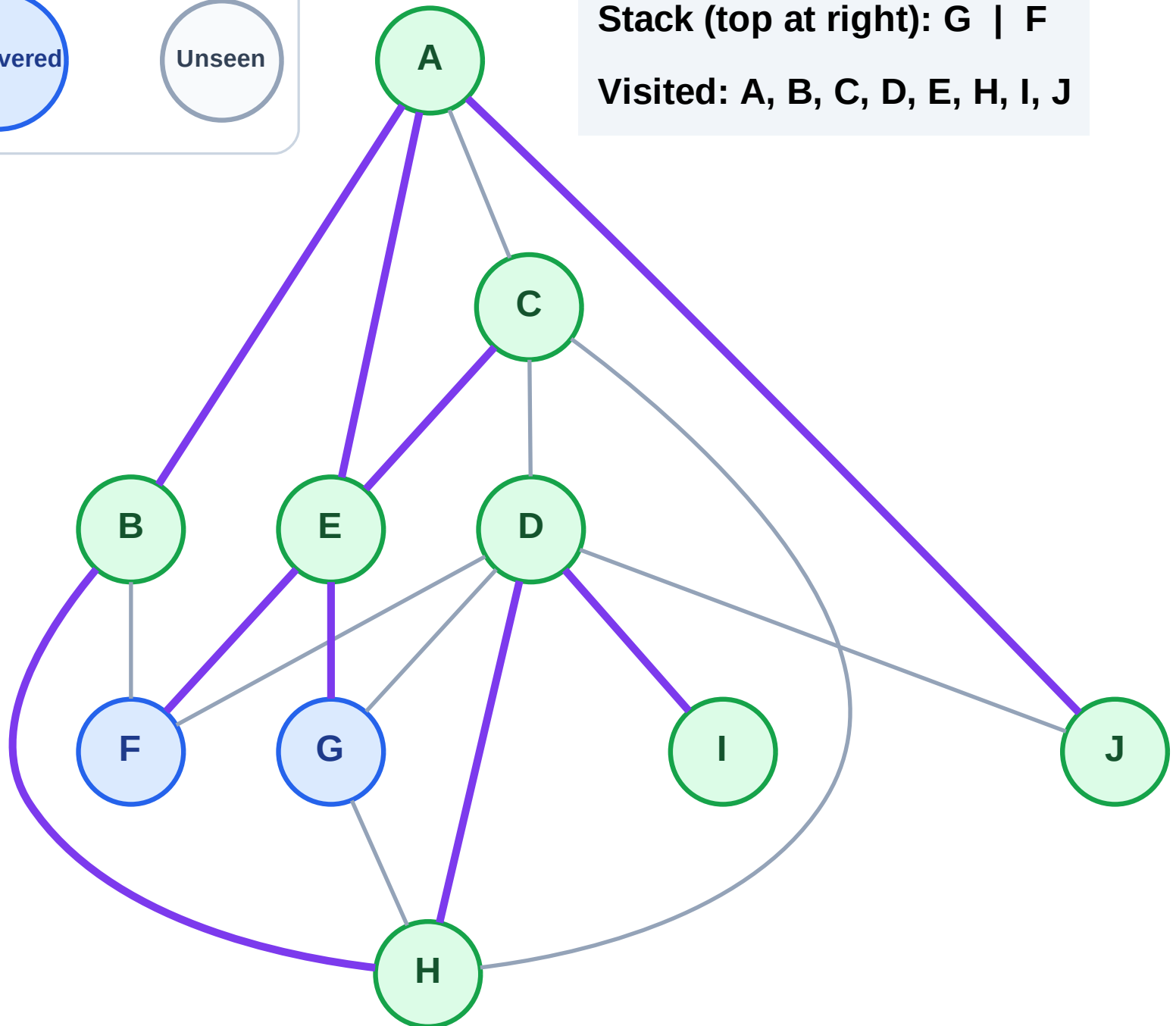
Step 17: No new neighbors from C

Legend



Stack (top at right): G | F

Visited: A, B, C, D, E, H, I, J



DFS Traversal

Step 18: Process node F

Legend

Complete

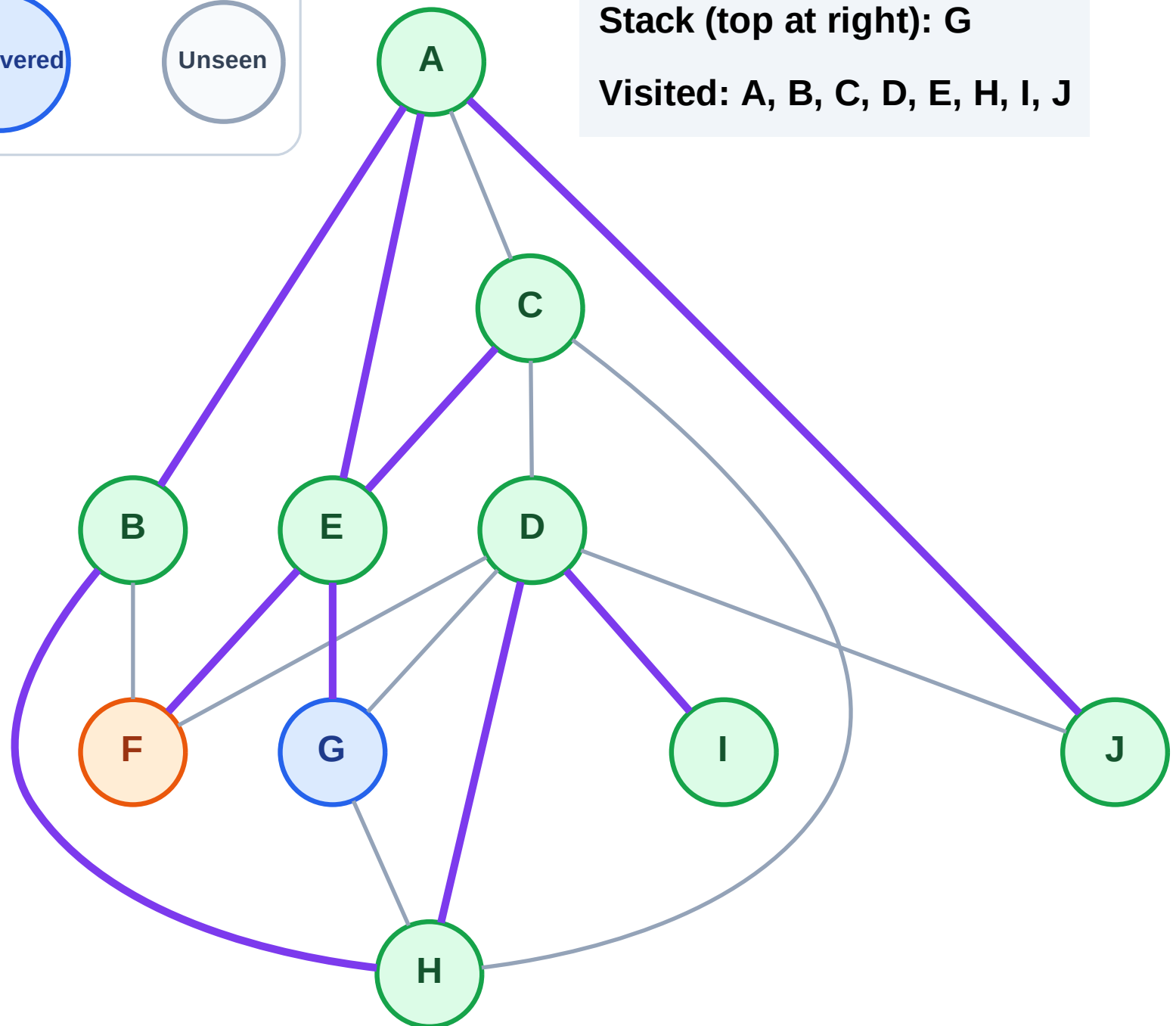
Processing

Discovered

Unseen

Stack (top at right): G

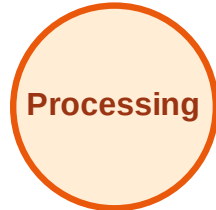
Visited: A, B, C, D, E, H, I, J



DFS Traversal

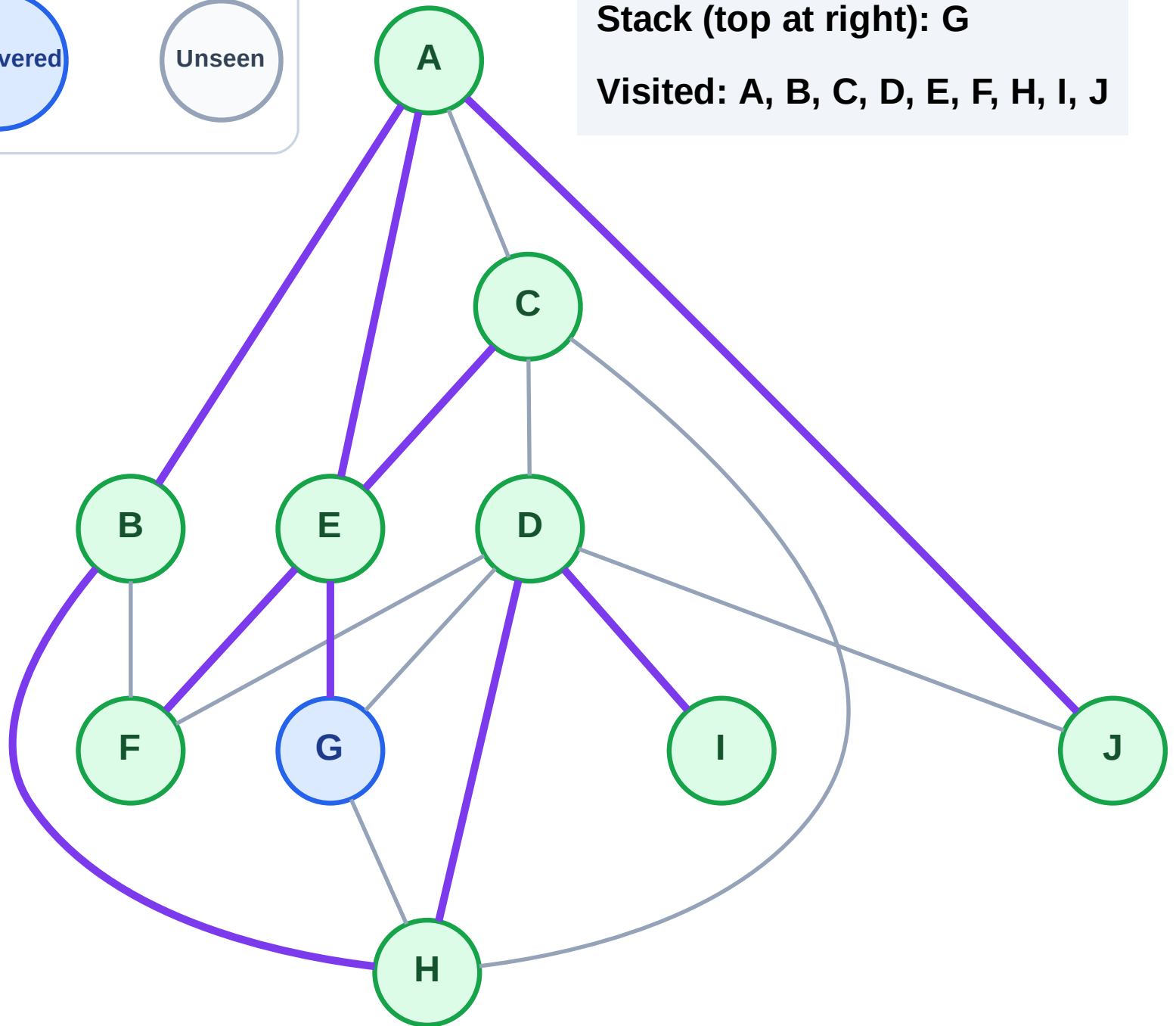
Step 19: No new neighbors from F

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

Complete

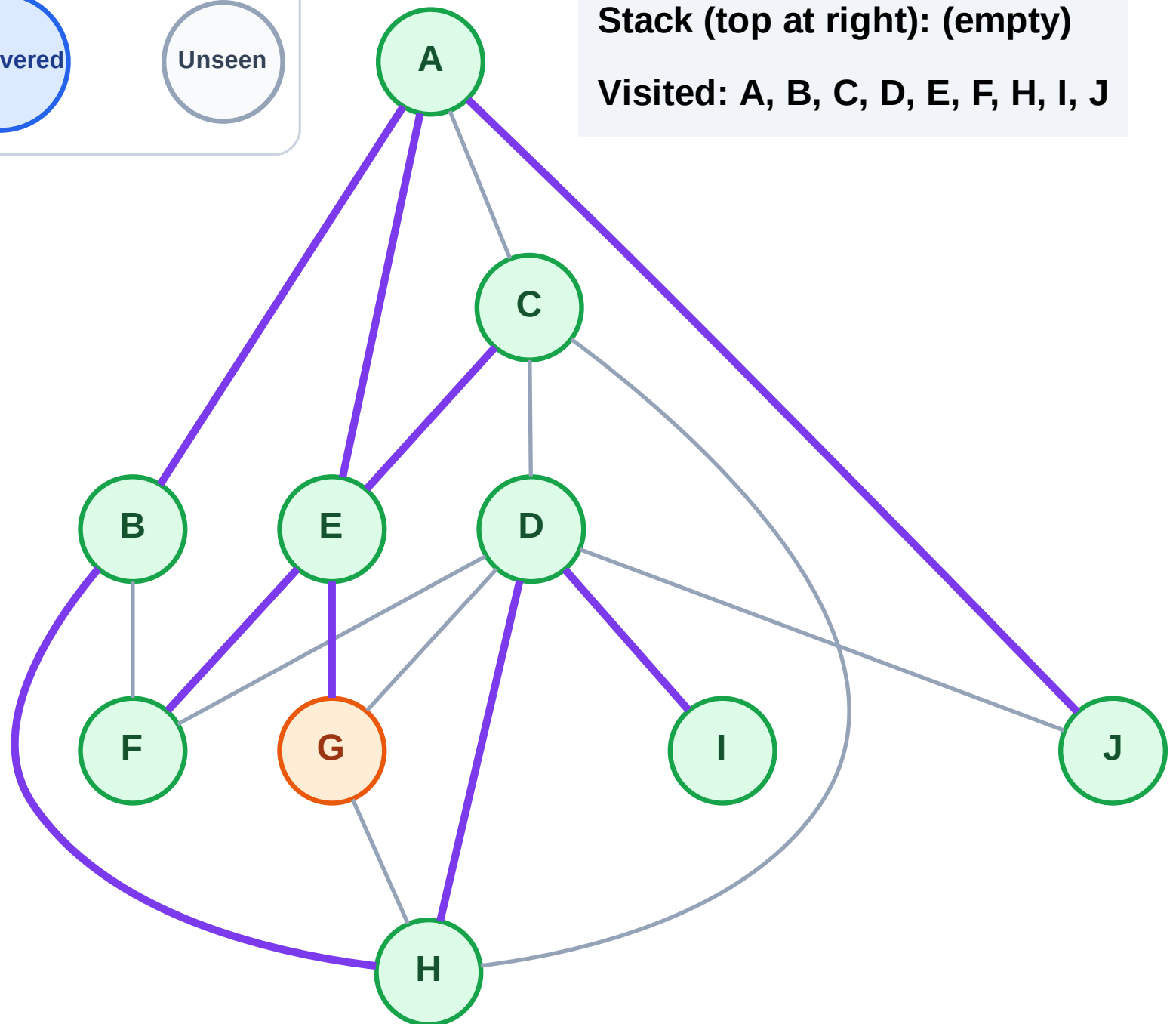
Processing

Discovered

Unseen

Stack (top at right): (empty)

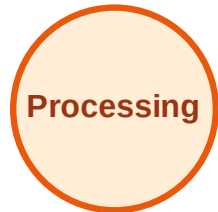
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

